

# Bilateral Bargaining with a Biased Intermediary \*

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## Abstract

Bilateral bargaining is often mediated by an *intermediary*. In many cases, an intermediary shares some interest with one of the two parties in the negotiation and cannot commit to and enforce his decision. This paper studies the implication of having such a biased intermediary without commitment and enforcement power on bargaining outcome. I consider a bilateral trade model à la [Myerson and Satterthwaite \(1983\)](#) with binary valuations in which the intermediary offers a price to the traders. I focus on the set of communication equilibria of the game, which characterizes what the players can achieve if they are allowed to make any preplay and intraplay communication. I show that biased mediation makes it harder to achieve ex post efficiency. I also characterize the second-best equilibrium and show that the expected social surplus is weakly decreasing in the degree of the intermediary's bias. The comparison with the seller-offer bargaining game shows that if the degree of bias is small enough, even a biased intermediary is useful in the second-best case.

**Keywords:** Bargaining, intermediary, bias, mediator, communication equilibrium.

**JEL Classification:** C78, D82.

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# 1 Introduction

Bilateral bargaining is often mediated by an *intermediary*, who communicates with the two parties in the negotiation and offers them a resolution. Such mediation obfuscates the private information inherent in offers and counter-offers that would be directly revealed absent an intermediary. For this reason, mediation is believed to mitigate strategic incentives in communication and help achieve efficient outcomes.

In many cases, however, an intermediary is not benevolent and shares some interest with one of the two parties in the negotiation. He opportunistically behaves and tries to achieve the outcomes he desires, which would harm efficiency of the bargaining outcome if his objective is not aligned with it.<sup>1</sup> He is not omnipotent either in the sense that he cannot commit to and enforce his decision. The lack of commitment and enforcement power hinders his ability to mediate the bargaining. For example, consider a real estate agent. He facilitates transactions by communicating with a seller and a buyer and offering them a price. Since he gets a portion of a sale price as a commission, he has the incentive to realize a higher sale price, being *biased* toward the seller.<sup>2</sup> Typically, he does not commit to the price he offers in advance, and trade occurs only when the seller and the buyer accept his offer.

The purpose of this paper is to study the implication of having a *biased intermediary without commitment and enforcement power* on bargaining outcome. To draw a connection to the mechanism design literature, I study a game played by a seller, a buyer, and an intermediary in a bilateral trade setting with binary valuations that is reminiscent of [Myerson and Satterthwaite \(1983\)](#). The seller owns a good and wants to trade it with the buyer. Each trader's valuation of the good is his private information and is independently and binary distributed. There is always gain from trade except for the pair of high type seller and low-type buyer. If there is trade, the seller's payoff is the price minus his valuation and the commission to the intermediary; the buyer's payoff is his valuation minus the price and the commission to the intermediary; the intermediary's payoff is the sum of the commissions paid by the traders. If there is no trade, all players get payoffs

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<sup>1</sup>I use "he" for the intermediary. "She" is reserved for a mediator, who shortly appears.

<sup>2</sup>In the U.S., it has been standard that the seller pays the commissions to his agent *and* the buyer's agent; he pays 5% of a sale price as a commission, which is equally split between the two agents. The rule has recently changed so that a seller can choose not to pay for a buyer's agent and that a buyer needs to negotiate with his agent about the commission (see [Kamin \(2024\)](#)).

of zero. The *underlying game* proceeds as follows: first, the seller and the buyer observe the realizations of their types (valuations); second, the intermediary offers a price to the traders; and last, the traders simultaneously respond to the offer by either acceptance or rejection. If both traders accept it, trade occurs at that price; if at least one of them rejects it, no trade occurs. I am interested in what the three players can achieve in equilibria of the underlying game *if they are allowed to make any preplay and intraplay communication*.

To obtain the bound on the achievable outcomes, I augment the game by adding a *fictitious* mediator, who collects type reports from the traders and recommends an action to each player. The *mediated bargaining game* proceeds as follows: first, the mediator publicly commits to a *mediation plan*, a function from type reports to recommendations; second, the seller and the buyer observe the realizations of their types and privately report them to the mediator; third, the mediator privately recommends a price to the intermediary, and the mediator observes it; fourth, the intermediary offers a price to the traders; fifth, the mediator privately recommends the traders' responses to the offer; and last, the traders simultaneously respond to the offer by either acceptance or rejection. If both traders accept it, trade occurs at that price; if at least one of them rejects it, no trade occurs.

The set of incentive-compatible mediation plans, known as the set of *communication equilibrium* (CE) due to Forges (1986), characterizes the set of achievable outcomes in the underlying game with some preplay and intraplay communication. In my model, however, some CEs are not robust to the players' trembles. Thus, I introduce a refinement of *acceptable CE* and use it as the equilibrium concept. Roughly speaking, acceptable CE requires that the mediator always recommends weakly dominant responses—accept an offer if and only if it guarantees a nonnegative payoff and reject it otherwise.

The first main result is that the biased mediation makes it harder to achieve ex post efficiency.<sup>3</sup> Specifically, I derive a necessary and sufficient condition under which ex post efficiency can be achieved in acceptable CE. This condition is more stringent than (i) when the intermediary does not take commissions and is solely interested in maximizing the probability of trade (*unbiased* intermediary) and (ii) when there is no intermediary (a mediated seller-offer bargaining game).<sup>4</sup>

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<sup>3</sup>Bargaining outcome is *ex post efficient* if trade occurs if and only if the buyer has a higher valuation.

<sup>4</sup>If an unbiased intermediary does take commissions, this is not necessarily the case. See Section C.1.2 for the

To prove this result, I first obtain a necessary condition for acceptable CE, which requires that the mediator always recommends the intermediary to offer either of the buyer's highest acceptable prices (one for each type). Intuitively, given that any price that guarantees both traders nonnegative payoffs will be accepted, the intermediary has the incentive to offer either of the buyer's highest acceptable prices. Combined with the definition of ex post efficiency, this necessary condition pins down a candidate mediation plan for an ex post efficient acceptable CE. Verifying the players' IC constraints, I show that ex post efficiency can be achieved if and only if the prior probability of the high-type buyer is small enough. This is because ex post efficiency requires the intermediary to offer the lower of the buyer's highest acceptable prices for some reports; to deter (i) the intermediary from deviating to the higher one, which will be accepted only by the pair of high type traders, and (ii) the low-type seller from misreporting his type, they must believe that the buyer is likely enough to be of low type. In contrast, if the intermediary is unbiased, he can be incentivized to offer prices different from the buyer's highest acceptable prices, which in general, enlarges the possibility for ex post efficiency. In the mediated seller-offer bargaining game, a necessary condition for acceptable CE identifies the same mediation plan as before as a candidate for an ex post efficient acceptable CE. Thus, ex post efficiency can be achieved if and only if the prior probability of the high-type buyer is small enough. The threshold is larger in the seller-offer bargaining game because the traders no longer pay the commissions to the intermediary.

As the second main result, I characterize the second-best outcome and show that a smaller bias is socially desirable because the expected social surplus is weakly decreasing in the degree of bias. The aforementioned necessary condition for acceptable CE implies that the players' IC constraints can be written as linear inequalities in the probabilities the mediator recommends one of the buyer's highest acceptable prices for each report. As such, the SB outcome can be characterized by solving the corresponding linear program. The solution to the program implies that there is a threshold ratio of the players' net payment/revenue at which the SB outcome changes its regime. The net payment/revenue ratio is increasing in the commission rates, hence interpreted as representing the degree of the intermediary's bias. Suppose the ratio is below the unbiased case and Section C.2 for the case of no intermediary.

threshold. In that case, in the SB acceptable CE, the intermediary offers the lower of the buyer's highest acceptable price with some probability, sacrificing the probability of trade between the high-type traders. The total probability of trade is decreasing in the ratio, so is the expected social surplus. If the ratio exceeds the threshold, however, the intermediary can no longer be incentivized to offer the lower price; he always offers the higher price in the SB acceptable CE. A simple calculation shows that the expected social surplus is higher in the former SB outcome. In the seller-offer bargaining game, the high-type seller cannot be incentivized to offer the lower price—he has no incentive to sacrifice the probability of trade against the high-type buyer. Thus, in the SB acceptable CE, the seller always offers the higher price, achieving exactly the same outcome as in the case of a biased intermediary when the ratio is below the threshold. This implies that if the intermediary's bias is small enough, having a biased intermediary leads to a higher expected social surplus than having no intermediary; even a biased intermediary can be useful in the SB case.

## 1.1 Related literature

Three features of the intermediary—bias, no commitment, and no enforcement—distinguish the present paper from other work in the literature. Intermediary, broadly defined, has been studied in the literature on mechanism design.<sup>5</sup> Classic papers such as [Myerson and Satterthwaite \(1983\)](#) and [Matsuo \(1989\)](#) can be considered studying the bargaining with an unbiased intermediary with commitment and enforcement power;<sup>6</sup> the principal maximizes the expected surplus; she can commit to a mechanism; and the agents' acceptance is not necessary once they participate in a mechanism.<sup>7</sup> The above three features have been separately studied in the literature. Papers that study optimal mechanism design by the seller, such as [Myerson \(1981\)](#), can be considered an extreme case of a biased intermediary; his preference is completely aligned with the seller. Also, [Myerson \(1982\)](#) studies a mechanism design problem when both hidden information and hidden action exist, which can be viewed as the principal having no enforcement power. Regarding the commitment assumption, there is a growing literature on mechanism design with limited

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<sup>5</sup>An intermediary in this literature is typically called a *principal*.

<sup>6</sup>The relationship with [Matsuo \(1989\)](#) is discussed in Appendix C.1.

<sup>7</sup>In this sense, “no enforcement” is equivalent to imposing ex post IR constraint instead of ex ante ones.

commitment (Bester and Strausz, 2001, 2000, 2007; Doval and Skreta, 2022; Lomys and Yamashita, 2021). Relatedly, Eilat and Pauzner (2021) study bilateral trade with a benevolent intermediary without commitment power.<sup>8</sup> Thus, the novelty of this paper is that the above three features are simultaneously present.<sup>9</sup> Combined with no commitment, the intermediary's bias gives him the incentive to offer as high a price as possible, which is the driving force of the results.

Biased intermediary has been studied in the context of international relations (Kydd, 2003, 2006). Kydd (2003) shows that an intermediary is effective in reducing the probability of conflict only if he is biased. Roughly speaking, this is because in his model the intermediary needs to credibly communicate his private information to avoid conflict. He shows that such credible communication is possible only for a biased intermediary. In contrast to his result, a biased intermediary does not help achieve ex post efficiency in the present paper. The crucial difference is that the traders do not need to know the opponent's type to achieve ex post efficiency. It is also worth mentioning that Kydd (2003) considers a fixed communication protocol, whereas the present paper allows for any preplay and intraplay communication between the players.<sup>10</sup>

The present paper is also distinct from some previous work in the literature in that the intermediary is an active player of the game, who has a preference and actions to choose from. This clarifies the effect of the intermediary's bias on his facilitative role (facilitate agreement by obfuscating private information inherent in offers and counter-offers).<sup>11</sup> See Čopič and Ponsatí (2008); Fanning (2021, 2023); Jarque, Ponsatí and Sákovics (2003) for the facilitative role and also Ganguly and Ray (2012); Goltsman, Hörner, Pavlov and Squintani (2009); Hörner, Morelli and Squintani (2015) for other studies on mediation.

Methodologically, I lean on the work of Forges (1986), Myerson (1986b), and Sugaya and Wolitzky (2021), who study multistage games with communication and prove the *communication revelation principle* for various equilibrium concepts. The communication revelation principle establishes that any equilibrium outcome of the underlying game with some preplay and intraplay

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<sup>8</sup>They directly study the game between the traders and the intermediary, in which the intermediary offers a mechanism. Note that the intermediary in my model just offers a price and is not allowed to offer a mechanism. In this regard, Eilat and Pauzner (2021) allow more flexibility in the intermediary's actions.

<sup>9</sup>Some papers combine the two of the abovementioned features. For example, Lomys and Yamashita (2021); Doval and Skreta (2024) study the seller-optimal mechanisms assuming limited commitment.

<sup>10</sup>The relationship with Kydd (2003) is also discussed in Section 5.2.2.

<sup>11</sup>Gottardi and Mezzetti (2024) study not only on this facilitative role but also on the evaluative role of the intermediary; the intermediary can provide some guidance on the appropriate resolution based on his expertise.

communication is achievable as a CE of the mediated bargaining game. In particular, it is without loss of generality to focus on the direct communication between the players and the mediator in the mediated bargaining game, as described above.

The remainder of the paper is organized as follows. Section 2 describes the model and define equilibrium concepts. In Section 3, I present some preliminary results. Section 4 studies the bound on the expected social surplus achievable in equilibrium. In particular, I derive a necessary and sufficient condition for achieving ex post efficiency and also characterize the second-best outcome. In Section 5, I compare the results in Section 4 with those in the seller-offer bargaining game. I also briefly discuss the implications of other payoff specifications and their relationship with the literature. Section 6 concludes the paper. All proofs are relegated to Appendix A.

## 2 Model

There are three players: a seller, a buyer, and an intermediary. The seller owns a good and wants to trade it with the buyer. The traders' valuations of the good are binary and independently distributed, and their realizations are private information of the respective trader. Specifically, the seller's valuation of the good is high ( $s_H$ ) with probability  $\pi_S \in (0, 1)$  and low ( $s_L$ ) with probability  $1 - \pi_S$ . The buyer's valuation of the good is high ( $b_H$ ) with probability  $\pi_B \in (0, 1)$  and low ( $b_L$ ) with probability  $1 - \pi_B$ . Let  $\Theta_S = \{s_L, s_H\}$  and  $\Theta_B = \{b_L, b_H\}$  be the set of possible valuations of the seller and the buyer, respectively. With slight abuse of notation, I denote the seller with valuation  $s \in \Theta_S$  (resp. the buyer with valuation  $b \in \Theta_B$ ) by  $s$  (resp.  $b$ ). I assume that there is always gain from trade except for the pair of traders  $(s_H, b_L)$ ; that is,  $0 < s_L < b_L < s_H < b_H$ . If  $(s, b)$  trade at price  $p$ , the seller gets  $(1 - \delta_S)p - s$ , the buyer gets  $b - (1 + \delta_B)p$ , and the intermediary gets  $(\delta_S + \delta_B)p$ , where  $\delta_S, \delta_B \in (0, 1)$ ; if there is no trade, all the players get 0. This payoff specification is interpreted as each trader paying the commissions  $\delta_S p$  and  $\delta_B p$  to the intermediary.<sup>12</sup> Since the intermediary's payoff is increasing in price  $p$ , he shares some interest with the seller. In this sense, he is considered *biased* toward the seller.<sup>13</sup>

The *underlying game* I consider here proceeds as follows: (i) the seller and the buyer observe

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<sup>12</sup>I discuss alternative specifications in Section 5.

<sup>13</sup>If the intermediary's payoff is independent of price (for example, if he gets a fixed commission from trade), he is considered *unbiased*. I explore this case in Appendix C.1

the realizations of their types (valuations); (ii) the intermediary offers a price to the traders; (iii) the traders simultaneously respond to the offer. If both traders accept it, trade occurs at that price; if at least one of them rejects it, no trade occurs. I am interested in what the three players can achieve in the underlying game *if they can freely communicate with each other at any point in the game* (sending messages, tossing a coin, using a correlation device, etc.) However, there are infinitely many possible forms of preplay and intraplay communication between the players, hence infinitely many games I need to consider. To circumvent this issue, I consider a mediated version of the underlying game, which is the subject of Section 2.1.

## 2.1 The mediated bargaining game

I augment the model by adding a *fictitious* mediator, who collects type reports from the traders between (i) and (ii) and later privately recommends action to each player. She carries out such mediation using a decision rule called a mediation plan. A *pure mediation plan* is a pair of functions  $(q, r)$ , where  $q: \Theta_S \times \Theta_B \rightarrow \mathbb{R}_+$  is a *price-recommendation* and  $r: \Theta_S \times \Theta_B \times \mathbb{R}_+^2 \rightarrow \{Y, N\}^2$  is a *response-recommendation*.<sup>14</sup> For example, suppose that the mediator uses a pure mediation plan  $(q, r)$  and that the traders have reported  $(\tilde{s}, \tilde{b}) \in \Theta_S \times \Theta_B$ . Then, the mediator recommends the intermediary to offer  $p = q(\tilde{s}, \tilde{b}) \in \mathbb{R}_+$ . After observing the intermediary's offer  $\tilde{p}$ , she recommends  $r(\tilde{p} \mid \tilde{s}, \tilde{b}, p) = (r_S(\tilde{p} \mid \tilde{s}, \tilde{b}, p), r_B(\tilde{p} \mid \tilde{s}, \tilde{b}, p)) \in \{Y, N\}^2$  to the traders, where  $r_S$  is the Seller's response (acceptance "Y" or rejection "N") to the offer and  $r_B$  is the Buyer's one.<sup>15</sup> Note that even though  $r(\tilde{p} \mid \tilde{s}, \tilde{b}, p)$  specifies both traders' responses, the mediator privately recommends each trader's response; trader  $i \in \{S, B\}$  only observes  $r_i(\tilde{p} \mid \tilde{s}, \tilde{b}, p) \in \{Y, N\}$ . Let  $Q$  and  $R$  be the set of all price-recommendations and response-recommendations, respectively. For greatest generality, I allow the mediator to use a *mixed mediation plan* (or simply a *mediation plan*), a probability distribution  $\mu \in \Delta(Q \times R)$ , where  $\Delta(X)$  denote the set of all probability distributions over the set  $X$ . Let  $\mu(q, r)$  denote the probability that the mediator uses pure mediation rule  $(q, r)$ .

<sup>14</sup>Strictly speaking, the communication revelation is established only for finite games. Hence, the intermediary's action space (the set of possible prices) must be finite to ensure the validity of the entire analysis. If I assume a finite set of possible prices, I can obtain qualitatively the same results. For the brevity of the exposition, I use the current continuous formulation.

<sup>15</sup>I denote  $r(\tilde{s}, \tilde{b}, p, \tilde{p})$  by  $r(\tilde{p} \mid \tilde{s}, \tilde{b}, p)$ .

I consider the *mediated bargaining game* that proceeds as follows:

1. The mediator publicly commits to a mediation plan  $\mu \in \Delta(Q \times R)$  she uses;
2. The seller and the buyer privately observe the realizations of their types;
3. The traders privately report their types  $\tilde{s} \in \Theta_S$  and  $\tilde{b} \in \Theta_B$  to the mediator;
4. The mediator privately picks up a pure mediation plan  $(q, r) \in Q \times R$  with probability  $\mu(q, r)$  and then privately recommends a price  $p = q(\tilde{s}, \tilde{b}) \in \mathbb{R}_+$  to the intermediary;
5. The intermediary offers a price  $\tilde{p} \in \mathbb{R}_+$  to the traders, and the mediator observes it;
6. The mediator privately recommends traders' responses  $r(\tilde{p} \mid \tilde{s}, \tilde{b}, p) \in \{Y, N\}^2$ ;
7. The traders simultaneously respond to the offer  $\tilde{p}$  by either acceptance ( $Y$ ) or rejection ( $N$ ). If both traders accept it, trade occurs at the price  $\tilde{p}$ , and the payoffs are realized; if at least one of them rejects it, no trade occurs, and all players get a payoff of 0.

**Remark 1.** As I discussed in Introduction, on top of bias, the intermediary features no commitment and no enforcement. Indeed, he does not commit to the price he offers; he updates his belief about the traders' types upon receiving a recommendation and offers an optimal price for him given his belief. Moreover, the traders' acceptances are necessary for trade to occur at the offered price. In contrast to the intermediary, the mediator *can* commit to the mediation plan she uses.

## 2.2 Communication equilibrium and its refinement

Each player can manipulate a mediation plan by disobeying recommendations, and the traders can also manipulate it by misreporting their types. In an equilibrium mediation plan, which is defined shortly, none of such manipulations are profitable.

I use *communication equilibrium* (CE) of Forges (1986) as an equilibrium concept. In CE, no player could ex ante expect to gain by manipulation (disobeying and misreporting). In other words, no player could expect to gain by manipulation after any event that is observable to him and has a *positive probability* of occurring in equilibrium.<sup>16</sup>

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<sup>16</sup>Myerson (1986b) considers a refinement of CE, *sequential communication equilibrium* (SCE). SCE requires

It is worth noting that there is no loss of generality coming from the form of communication in the mediated bargaining game (namely, Time 3, 4, and 6). In principle, I can consider more complex forms of communication between the players and the mediator. For example, the traders' message space can be larger than their type spaces, and the mediator can communicate more than recommended actions to the players. However, thanks to the communication revelation principle shown by [Forges \(1986\)](#), [Myerson \(1986b\)](#) and, [Sugaya and Wolitzky \(2021\)](#), any equilibrium outcome of the underlying game with some preplay and intraplay communication is achievable as a CE of the mediated bargaining game. As such, it suffices to characterize the set of CE outcomes.

To formally define CE, let  $V(q, r)$  be the intermediary's ex ante expected payoff when the mediator uses a pure mediation plan  $(q, r)$  and all players are honest and obedient to her:

$$V(q, r) = \sum_{(s, b) \in \Theta_S \times \Theta_B} \Pr(s, b) (\delta_S + \delta_B) q(s, b) \mathbf{1}_{\{r(q(s, b) | s, b, q(s, b)) = (Y, Y)\}},$$

where  $\Pr(s, b)$  is the prior probability that the traders' types are  $(s, b)$ , and  $\mathbf{1}_{\{\cdot\}}$  is the indicator function. Similarly, for trader  $i \in \{S, B\}$ , let  $U_i(q, r)$  be  $i$ 's ex ante expected payoff when the mediator uses a pure mediation plan  $(q, r)$  and all players are honest and obedient to her:

$$\begin{aligned} U_S(q, r) &= \sum_{(s, b) \in \Theta_S \times \Theta_B} \Pr(s, b) [(1 - \delta_S) q(s, b) - s] \mathbf{1}_{\{r(q(s, b) | s, b, q(s, b)) = (Y, Y)\}}, \\ U_B(q, r) &= \sum_{(s, b) \in \Theta_S \times \Theta_B} \Pr(s, b) [b - (1 + \delta_B) q(s, b)] \mathbf{1}_{\{r(q(s, b) | s, b, q(s, b)) = (Y, Y)\}}. \end{aligned}$$

Manipulation by the intermediary can be represented by a function  $\alpha: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ . Let  $\Sigma_I$  be the set of all such functions. Manipulation by trader  $i \in \{S, B\}$  can be represented by a pair of functions  $(\beta_i, \gamma_i)$ , where  $\beta_i: \Theta_i \rightarrow \Theta_i$  is a manipulation by report, and  $\gamma_i: \Theta_i \times \mathbb{R}_+ \times \{Y, N\} \rightarrow \{Y, N\}$  is a manipulation by response. For  $i \in \{S, B\}$ , let  $\Sigma_i$  be the set of all such pairs of functions.

For any  $\alpha \in \Sigma_I$ , let  $\alpha \circ q$  be such that  $(\alpha \circ q)(s, b) = \alpha(q(s, b))$  for all  $(s, b) \in \Theta_S \times \Theta_B$ ;

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there being no profitable manipulation even after events that have *zero-probability* of occurring in equilibrium. As I show in [Appendix B.1](#), every CE is an SCE in the mediated bargaining game. Thus, I work with CE in the remainder of the paper.

that is,  $\alpha \circ q$  gives the intermediary's offer as a function of reports  $(s, b)$  when the mediator uses  $q$  and the intermediary manipulates it by  $\alpha$ . For any  $(\beta_i, \gamma_i) \in \Sigma_i$ , let  $q \circ \beta_i$  and  $r \circ (\beta_i, \gamma_i)$  be such that, for all  $(\theta_i, \theta_j) \in \Theta_i \times \Theta_j$  and  $p \in \mathbb{R}_+$ ,<sup>17</sup>

$$(q \circ \beta_i)(\theta_i, \theta_j) = q(\beta_i(\theta_i), \theta_j)$$

$$(r \circ (\beta_i, \gamma_i))(p \mid \theta_i, \theta_j) = \left( \gamma_i(\theta_i, p, r_i(p \mid \beta_i(\theta_i), \theta_j, q(\beta_i(\theta_i), \theta_j))), r_j(p \mid \beta_i(\theta_i), \theta_j, q(\beta_i(\theta_i), \theta_j)) \right)$$

where  $j \neq i$ ; that is,  $q \circ \beta_i$  gives the intermediary's offer when the mediator uses  $q$ , trader  $i$  manipulates it by  $\beta_i$ , and the other trader  $j$  and the intermediary are honest and obedient to the mediator. Similarly,  $r \circ (\beta_i, \gamma_i)$  gives the traders' responses when the mediator uses  $(q, r)$ , trader  $i$  manipulates it by  $(\beta_i, \gamma_i)$ , and the other trader  $j$  and the intermediary are honest and obedient to the mediator. To simplify the notation, let  $(q, r) \circ (\beta_i, \gamma_i)$  denote  $(q \circ \beta_i, r \circ (\beta_i, \gamma_i))$ . For any  $\alpha \in \Sigma_I$  and  $(\beta_i, \gamma_i) \in \Sigma_i$ , the ex ante expected payoffs of the intermediary and trader  $i$  when they use these manipulations,  $V(\alpha \circ q, r)$  and  $U_i((q, r) \circ (\beta_i, \gamma_i))$ , respectively, are naturally defined. With these notations, I can now define CE.

**Definition 1.** A mediation plan  $\mu \in \Delta(Q \times R)$  is a *communication equilibrium (CE)* if, for all  $\alpha \in \Sigma_I$ ,

$$\sum_{(q, r) \in Q \times R} \mu(q, r) V(q, r) \geq \sum_{(q, r) \in Q \times R} \mu(q, r) V(\alpha \circ q, r), \quad (2.1)$$

and for all  $i \in \{S, B\}$  and  $(\beta_i, \gamma_i) \in \Sigma_i$ ,

$$\sum_{(q, r) \in Q \times R} \mu(q, r) U_i(q, r) \geq \sum_{(q, r) \in Q \times R} \mu(q, r) U_i((q, r) \circ (\beta_i, \gamma_i)). \quad (2.2)$$

**Remark 2.** In Appendix B.2, I show that the *ex ante* IC constraints (2.1) and (2.2) are satisfied if and only if the *interim* IC constraints are satisfied; that is, (i) no trader can gain by manipulation *after* he learns his type (at Time 3) and (ii) the intermediary cannot gain by manipulation *after* he receives a recommendation (at Time 5). Accordingly, I consider the players' interim IC

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<sup>17</sup>Abusing the notation, I write  $r \circ (\beta_i, \gamma_i)$  instead of  $(q \circ r) \circ (\beta_i, \gamma_i)$  as I use the latter to denote the pair  $(q \circ \beta_i, r \circ (\beta_i, \gamma_i))$ .

constraints in the remainder of the paper as they are easier to handle than the ex ante ones.

### 2.2.1 Acceptable CE

Without any restriction on admissible response-recommendations, it is easy to construct a CE that implements an ex post efficient outcome since the mediator can punish the intermediary's deviations. Indeed, consider a mediation rule that, whenever the buyer has a higher valuation, recommends (i) the intermediary to offer a mutually acceptable price (a price that guarantees nonnegative payoffs to both traders) and (ii) the traders to accept an offer only when the offer is mutually acceptable and the intermediary has not deviated. Such a mediation plan can constitute a CE under certain condition. However, such CEs are not robust to the players' trembles. For example, suppose the intermediary, by mistake, offers a mutually acceptable price different from the recommended one. In that case, the traders can be better off by accepting this offer though they are recommended to punish the intermediary by rejecting it.

Such non-robust CEs can be eliminated if the traders' are always recommended to accept mutually acceptable prices, which is reminiscent of a weakly dominant strategy. To formally define such response-recommendation, observe that the seller with valuation  $s$  gets a nonnegative payoff by accepting  $p$  if and only if  $p \geq \underline{p}_s \equiv \frac{s}{1-\delta_S}$ . Similarly, the buyer with valuation  $b$  gets a nonnegative payoff by accepting  $p$  if and only if  $p \leq \overline{p}_b \equiv \frac{b}{1+\delta_B}$ . I assume  $\underline{p}_{s_L} \leq \overline{p}_{b_L}$  and  $\underline{p}_{s_H} \leq \overline{p}_{b_H}$  so that there is room for achieving ex post efficiency.<sup>18</sup> Define  $r^A \in R$  as follows. For all  $p \in \mathbb{R}_+$  and all  $(s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\}$ ,

$$r^A(\tilde{p} \mid s, b, p) = \begin{cases} (N, Y) & \text{if } \tilde{p} \in [0, \underline{p}_s) \\ (Y, Y) & \text{if } \tilde{p} \in [\underline{p}_s, \overline{p}_b] \\ (Y, N) & \text{if } \tilde{p} \in (\overline{p}_b, +\infty) \end{cases}$$

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<sup>18</sup>These conditions are reduced to  $h \equiv \frac{1+\delta_B}{1-\delta_S} \leq \min \left\{ \frac{b_L}{s_L}, \frac{b_H}{s_H} \right\}$ . If this does not hold, at least one of  $(s_L, b_L)$ ,  $(s_L, b_H)$ , and  $(s_H, b_H)$  cannot agree to trade though there is gain from trade.

and for all  $p \in \mathbb{R}_+$ ,

$$r^A(\tilde{p} \mid s_H, b_L, p) = \begin{cases} (N, Y) & \text{if } \tilde{p} \in [0, \overline{p_{b_L}}] \\ (N, N) & \text{if } \tilde{p} \in (\overline{p_{b_L}}, \underline{p_{s_H}}) \\ (Y, N) & \text{if } \tilde{p} \in [\underline{p_{s_H}}, +\infty). \end{cases}$$

That is,  $r^A$  neglects the intermediary's deviations and recommends each trader to accept offer  $\tilde{p}$  if and only if it guarantees him a nonnegative payoff.<sup>19</sup> Thus, each trader finds it optimal to follow  $r^A$  if he has honestly reported his type. I restrict attention to CE that only uses  $r^A$  as a response-recommendation.

**Definition 2.** A CE  $\mu \in \Delta(Q \times R)$  is *acceptable* if  $\mu(Q \times \{r^A\}) = 1$ .<sup>20</sup>

### 3 Preliminary Results

In this section, I first provide a necessary condition for acceptable CE and use it to simplify the players' IC constraints.

It is straightforward to show that in any acceptable CE, the mediator recommends the intermediary to offer either  $\overline{p_{b_L}}$  and  $\overline{p_{b_H}}$  whenever there is gain from trade. To see this, suppose that the intermediary is recommended  $p \notin \{\overline{p_{b_L}}, \overline{p_{b_H}}\}$ . If  $p < \overline{p_{b_L}}$  and he expects it to be accepted, then he can gain by deviating to  $\overline{p_{b_L}}$  since it will also be accepted given the definition of  $r^A$ . Similarly, if  $\overline{p_{b_L}} < p < \overline{p_{b_H}}$  and he expects it to be accepted, then he can gain by deviating to  $\overline{p_{b_H}}$ . Formalizing this argument, I can show the following lemma.

**Lemma 1.** A mediation plan  $\mu \in \Delta(Q \times \{r^A\})$  is an acceptable CE only if it recommends the intermediary to offer either  $\overline{p_{b_L}}$  or  $\overline{p_{b_H}}$  whenever there is gain from trade. That is, for all  $q \in \text{supp}(\mu)$ ,

$$q(s, b) \in \{\overline{p_{b_L}}, \overline{p_{b_H}}\} \text{ for all } (s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\} \quad (3.1)$$

<sup>19</sup>Since  $r^A$  does not depend on the recommended price  $p$ , abusing the notation, I hereafter omit it from the arguments of  $r^A$ .

<sup>20</sup>This refinement has a flavor of *acceptable correlated equilibrium* of Myerson (1986a), hence the name.

*Proof.* See Appendix A.1. □

Lemma 1 implies that for any candidate for acceptable CE, the players' IC constraints can be written as functions of the total probability that the mediator recommends offering price  $\overline{p_{b_H}}$  for each pair of reports. For any mediation plan  $\mu \in \Delta(Q \times \{r^A\})$  that satisfies (3.1), let

$$\begin{aligned} x_{HH}(\mu) &= \sum_{q \in \{q' \in Q: q'(s_H, b_H) = \overline{p_{b_H}}\}} \mu(q), \\ x_{LH}(\mu) &= \sum_{q \in \{q' \in Q: q'(s_L, b_H) = \overline{p_{b_H}}\}} \mu(q), \\ x_{LL}(\mu) &= \sum_{q \in \{q' \in Q: q'(s_L, b_L) = \overline{p_{b_H}}\}} \mu(q), \end{aligned}$$

where  $\mu(q)$  denotes the probability that the mediator picks up pure mediation plan  $(q, r^A)$  when she uses  $\mu$ . For example,  $x_{HH}(\mu)$  is the total probability that  $\mu$  recommends the intermediary to offer  $\overline{p_{b_H}}$  when the reports are  $(s_H, b_H)$ . These probabilities completely describe the price-recommendations for all reports  $(s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\}$ . Note that by definition of  $r^A$ , traders  $(s_H, b_L)$  never trade if they are obedient to the mediator. Since the objective is to characterize the bound on the implementable outcomes, it suffices to recommend offering a high enough price for reports  $(s_H, b_L)$ . This deters traders from double deviations (misreport and then disobey) and makes the players' IC constraints easier to satisfy. For this reason, I restrict attention to price-recommendations in  $Q_x \subset Q$  defined below:

$$Q_x = \{q \in Q: q \text{ satisfies (3.1) and } q(s_H, b_L) \in (\overline{p_{b_H}}, +\infty)\}.$$

For any mediation plan  $\mu \in \Delta(Q \times \{r^A\})$  that satisfies (3.1) for all  $q \in \text{supp}(\mu)$ , I can construct another mediation plan  $\mu_x \in \Delta(Q_x \times \{r^A\})$  that has weakly less stringent IC constraints than  $\mu$  and induces the same distribution over the bargaining outcomes (whether trade occurs or not, and if it does, at which price) if the players are honest and obedient to the mediator. To explain how to construct it, for any  $q$  in the support of such  $\mu$ , let  $x_q \in Q_x$  be an arbitrary price-recommendation such that  $x_q(s, b) = q(s, b)$  for all  $(s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\}$ . That is,  $x_q$  recommends the same prices as  $q$  for all reports  $(s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\}$  but recommends a high

price in  $(\overline{p_{b_H}}, +\infty)$  for reports  $(s_H, b_L)$ . The mediation plan  $\mu_x$  can be constructed by letting  $\mu_x(x_q) = \mu(q)$  for all  $q \in \text{supp}(\mu)$ . Lemma 2 allows me to focus on such  $\mu_x$ .

**Lemma 2.** *For any mediation plan  $\mu \in \Delta(Q \times \{r^A\})$ , if  $\mu$  is an acceptable CE, so is  $\mu_x$ .*

*Proof.* See Appendix A.2. □

**Remark 3.** The proof of Lemma 2 implies that for any mediation plan in  $\Delta(Q_x \times \{r^A\})$  characterized by total probabilities  $x_{HH}$ ,  $x_{LH}$ , and  $x_{LL}$ , the players' IC constraints are given by the following linear inequalities in these probabilities:

$$x_{HH} \geq x_{LH} \tag{IC- $s_H$ }$$

$$x_{LL} \geq x_{LH} \tag{IC- $b_H$ }$$

$$(1 - \pi_B)(b_L - h s_L)(1 - x_{LL}) \geq \pi_B(b_H - b_L)(x_{HH} - x_{LH}) \tag{IC- $s_L$ }$$

$$\pi_S \pi_B b_H x_{HH} + (1 - \pi_S) \pi_B (b_H - b_L) x_{LH} \geq (1 - \pi_S)(1 - \pi_B) b_L x_{LL} \tag{IC-Int1}$$

$$(1 - \pi_S)(1 - \pi_B) b_L (1 - x_{LL}) \geq \pi_S \pi_B b_H (1 - x_{HH}) + (1 - \pi_S) \pi_B (b_H - b_L) (1 - x_{LH}) \tag{IC-Int2}$$

As such, if I am interested in a maximum (or a maximizer) of some objective function that is also linear in  $x_{HH}$ ,  $x_{LH}$ , and  $x_{LL}$ , it can be found by solving a corresponding linear program.

## 4 Bound on the Expected Social Surplus

In this section, I take the expected social surplus (the expected value of the sum of the players' payoffs) as an objective function to be maximized. Since the payments cancel out, the social surplus coincides with the trade surplus. Hence, for any  $\mu \in \Delta(Q_x \times \{r^A\})$  characterized by  $x_{HH}$ ,  $x_{LH}$ , and  $x_{LL}$ , the expected social surplus is given by

$$\begin{aligned} & \sum_{(s,b) \in \Theta_S \times \Theta_B} \Pr(s,b) \sum_{q \in Q_x} \mu(q)(b-s) \mathbf{1}_{\{r^A(q(s,b) | s,b)=(Y,Y)\}} \\ &= \pi_S \pi_B (b_H - s_H) x_{HH} + (1 - \pi_S) \pi_B (b_H - s_L) + (1 - \pi_S)(1 - \pi_B)(b_L - s_L)(1 - x_{LL}). \end{aligned} \tag{4.1}$$

Thus, the problem is reduced to

$$\begin{aligned} & \max_{x_{HH}, x_{LH}, x_{LL} \in [0,1]} \quad (4.1) \\ & \text{subject to} \quad (\text{IC-}s_H), (\text{IC-}b_H), (\text{IC-}s_L), (\text{IC-Int1}), \text{ and } (\text{IC-Int2}). \end{aligned} \quad (\text{P})$$

## 4.1 Ex post efficiency

I first explore when ex post efficiency can be achieved in equilibrium. A mediation plan is *ex post efficient* if, assuming the players are honest and obedient to the mediator, trade occurs if and only if the buyer has a higher valuation. If an ex post efficient mediation plan is an acceptable CE, it is called *ex post efficient* acceptable CE. Given the definition of  $r^A$ , a mediation plan  $\mu \in \Delta(Q_x \times \{r^A\})$  is ex post efficient if and only if  $x_{HH} = 1$  and  $x_{LL} = 0$  so that it always recommends  $\overline{p}_{b_H}$  to  $(s_H, b_H)$  and  $\overline{p}_{b_L}$  to  $(s_L, b_L)$ .<sup>21</sup> Moreover,  $x_{LH} = 0$  is also necessary for such a mediation plan to be an acceptable CE. Indeed, if  $x_{LH} > 0$  and hence price  $\overline{p}_{b_L}$  is offered to  $(s_L, b_H)$  with probability  $1 - x_{LH} < 1$ , the high-type buyer  $b_H$  can gain by misreporting his type. In that case, he can secure positive payoff  $b_H - (1 + \delta_B)\overline{p}_{b_L} = b_H - b_L$  with probability  $1 - \pi_S$  while he gets the same payoff only with smaller probability  $(1 - \pi_S)(1 - x_{LH})$  if he is honest and obedient to the mediator. I summarize these observations in the next lemma.

**Lemma 3.** *A mediation plan  $\mu \in \Delta(Q_x \times \{r^A\})$  is ex post efficient if and only if  $x_{HH} = 1$  and  $x_{LL} = 0$ . Moreover, such a mediation plan is an acceptable CE only if  $x_{LH} = 0$ .*

By Lemma 3, it suffices to check when  $(x_{HH}, x_{LH}, x_{LL}) = (1, 0, 0)$  satisfy the players' IC constraints. Note that the high-type seller  $s_H$  and both types of buyers have no incentive for manipulation; if they honestly report their types, they find it optimal to follow  $r^A$  by definition; even if they misreport their types, they cannot trade at better terms. However, the low-type seller  $s_L$  may have the incentive to misreport his type to realize trade at a higher price  $\overline{p}_{b_H}$ , which is possible if the buyer is of high type. To deter him from such manipulation, he must believe that the buyer is likely enough to be of low type  $b_L$ . That is, the prior probability  $\pi_B$  of the high-type buyer must be small enough. The intermediary must not have the incentive to disobey recommendations either. If he is recommended  $\overline{p}_{b_H}$ , he learns that the traders are  $(s_H, b_H)$ , and

<sup>21</sup>When there is no confusion, I write  $x_{HH}$ ,  $x_{LH}$ , and  $x_{LL}$  instead of  $x_{HH}(\mu_x)$ ,  $x_{LH}(\mu_x)$ , and  $x_{LL}(\mu_x)$ .

hence following the recommendation is optimal. However, if he is recommended  $\overline{p_{b_L}}$ , he may have the incentive to offer  $\overline{p_{b_H}}$  to realize trade at a higher price, which occurs only when the buyer is of high type. He does not have such an incentive if  $\pi_B$  is small enough. It turns out that the seller's threshold is more stringent, and I obtain the following proposition.

**Proposition 1.** *There exists an ex post efficient acceptable CE if and only if the prior probability  $\pi_B$  of the high-type buyer satisfies  $\pi_B \leq \overline{\pi_B} \equiv \frac{b_L - h s_L}{b_H - h s_L}$ , where  $h \equiv \frac{1 + \delta_B}{1 - \delta_S}$ .*

*Proof.* See Appendix A.3. □

It is intuitive that  $\overline{\pi_B}$  is decreasing in  $\delta_S$ ,  $\delta_B$ , and  $s_L$ ; the higher the commission rates or the seller's reservation value, the smaller the possibility of efficient trade. In this sense, the intermediary's bias, or desire to realize trade at a higher price, is detrimental to ex post efficiency.

## 4.2 Second-best outcome

In this section, I characterize the *second-best* (SB) outcome—implementable outcome that maximizes the expected social surplus when ex post efficiency cannot be achieved. Acceptable CE that implements the SB outcome can be obtained as the solution to the linear program (P) when  $\pi_B > \overline{\pi_B}$ . The next proposition gives the characterization.

**Proposition 2.** *There exists a threshold  $\overline{h}$  on the ratio  $h$  such that*

1. *if  $h \in (1, \overline{h}]$ , then in the SB outcome, traders  $(s_H, b_H)$  do not trade with positive probability while  $(s_L, b_H)$  and  $(s_L, b_L)$  trade at price  $\overline{p_{b_L}}$  with probability 1;*
2. *otherwise, in the SB outcome, traders  $(s_H, b_H)$  and  $(s_L, b_H)$  trade at price  $\overline{p_{b_H}}$  with probability 1 while  $(s_L, b_L)$  do not trade.*

*Proof.* See Appendix A.4. □

As shown in the proof of Proposition 1, if  $\pi_B > \overline{\pi_B}$  then  $(x_{HH}, x_{LH}, x_{LL}) = (1, 0, 0)$  do not satisfy the low-type seller's IC constraint (IC- $s_L$ ) since he can gain by misreporting his type. Given the expression of the expected social surplus (4.1), the SB is likely to be achieved by keeping  $x_{LH} = x_{LL} = 0$  and decreasing  $x_{HH}$  until (IC- $s_L$ ) binds—giving up some probability

of trade between  $(s_H, b_H)$ —to deter  $s_L$  from misreporting. Indeed, the SB is characterized by  $(x_{HH}^*, x_{LH}^*, x_{LL}^*) = (\bar{x}_{HH}, 0, 0)$  if  $h \leq \bar{h}$ , where  $\bar{x}_{HH}$  is the value of  $x_{HH}$  at which (IC- $s_L$ ) binds when  $x_{LH} = x_{LL} = 0$ . Note that  $\bar{x}_{HH}$  is decreasing in  $h$ ; if the low-type seller is honest and obedient to the mediator, his expected payoff is  $(1 - \delta_S)\bar{p}_{b_L} - s_L = \frac{b_L}{h} - s_L$ . As ratio  $h$  increases, misreporting becomes more attractive to him, hence a smaller  $\bar{x}_{HH}$ .<sup>22</sup> Decreasing  $x_{HH}$  from 1 to  $\bar{x}_{HH}$  is innocuous to all other constraints except for the intermediary's constraint (IC-Int2), which ensures that he follows the recommendation of offering  $\bar{p}_{b_L}$ . When  $\bar{p}_{b_L}$  is recommended, the smaller  $x_{HH}$ , the more he believes the traders are  $(s_H, b_H)$ , who do not accept price  $\bar{p}_{b_L}$ . Thus, he has the incentive to disobey and offer  $\bar{p}_{b_H}$  if  $x_{HH}$  is too small. Therefore,  $h$  must be small enough for  $\bar{x}_{HH}$  to be compatible with the lower bound implied by (IC-Int2).

If ratio  $h$  exceeds the threshold  $\bar{h}$ , then (IC- $s_L$ ) and (IC-Int2) cannot be satisfied at the same time at  $(x_{HH}, x_{LH}, x_{LL}) = (\bar{x}_{HH}, 0, 0)$  because the intermediary has the incentive to disobey and offer  $\bar{p}_{b_H}$ . In this case, the SB involves the mediator always recommending him to offer  $\bar{p}_{b_H}$ . That is, the SB is  $(x_{HH}^*, x_{LH}^*, x_{LL}^*) = (1, 1, 1)$ . The following argument is not rigorous but gives some intuition into this case. Suppose  $\pi_B$  is large enough. When  $\bar{p}_{b_L}$  is recommended, the intermediary has the incentive to follow the recommendation if the traders are likely to be  $(s_L, b_L)$ , who reject  $\bar{p}_{b_H}$ . The smaller the value of  $x_{LL}$ , the more he believes so. Indeed, the left-side of (IC-Int2) is decreasing in  $x_{LL}$ . By (IC- $b_H$ ), he has the largest incentive to follow the recommendation  $\bar{p}_{b_L}$  when  $x_{LL} = x_{LH}$ . However, substituting this into (IC-Int2), it can be satisfied only when  $x_{HH} = x_{LH} = x_{LL} = 1$  since  $\pi_B$  is large. In other words, the intermediary can no longer be incentivized to offer  $\bar{p}_{b_L}$ . Thus, in the only acceptable CE hence the SB, the mediator always recommends  $\bar{p}_{b_H}$ .

From Proposition 2, a simple calculation shows that the expected social surplus is (weakly) decreasing in  $h$ . On  $h \in (1, \bar{h})$ , it is decreasing as the trade probability  $\bar{x}_{HH}$  for the traders  $(s_H, b_H)$  is decreasing in  $h$ . If  $h < \bar{h}$  increases to  $h' > \bar{h}$  and the SB outcome changes its regime, the expected social surplus decreases.

**Corollary 1.** *The expected social surplus is (weakly) decreasing in  $h$  on  $(1, \bar{h}]$  and constant on  $[\bar{h}, \min\{\frac{b_L}{s_L}, \frac{b_H}{s_H}\}]$ .*

<sup>22</sup>His expected payoff from the best manipulation is also decreasing in  $h$ . but the equilibrium payoff decreases faster than it.

*Proof.* See Appendix A.5. □

## 5 Discussion

In Section 5.1, I compare the results in the main model with those in the seller-offer bargaining game, which are formally analyzed in Appendix C.2. I also briefly discuss alternative payoff specifications and their implications in Section 5.2.<sup>23</sup>

### 5.1 Relationship with the seller-offer bargaining game

In Appendix C.2, I study another mediated bargaining game where there is no intermediary, and the seller instead makes an offer. Since the traders no longer need to pay the commissions, if the traders  $(s, b)$  trade at price  $p$ , the seller gets  $p - s$ , and the buyer gets  $b - p$ . I show in Proposition 4 that an ex post efficient acceptable CE exists if and only if

$$\pi_B \leq \frac{b_L - s_L}{b_H - s_L}.$$

Since  $\frac{b_L - s_L}{b_H - s_L} \geq \overline{\pi_B}$  if the traders cannot achieve ex post efficiency themselves, a biased intermediary cannot achieve it either. Moreover, it can be the case that ex post efficiency can be achieved without an intermediary but not with a biased one. This implies that a biased intermediary is not just ineffective in achieving ex post efficiency; the traders can be better off without him.

I also show that if  $\pi_B > \frac{b_L - s_L}{b_H - s_L}$  so that ex post efficiency cannot be achieved, there is a unique acceptable CE, where the seller always offers price  $b_H$ . This is exactly the SB outcome in the main model when the net payment/revenue ratio exceeds the threshold ( $h > \bar{h}$ ). Therefore, if the intermediary's bias is sufficiently small so that  $h \leq \bar{h}$ , having a biased intermediary leads to a higher expected social surplus than having no intermediary. Thus, even if the intermediary is biased, his ability to obfuscate private information can be useful. This result justifies the widespread use of intermediaries in bargaining when it is common knowledge that intermediaries are biased.

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<sup>23</sup>The proofs are available upon request.

## 5.2 Alternative payoff specifications

### 5.2.1 Intermediary as a representation of market mechanism

Loertscher and Marx (2022) study a model of bilateral bargaining and model the market as a mechanism that maximizes the expected weighted average of firms' (sellers and buyers) payoffs. Motivated by their work, I can think of the intermediary playing a role of the market mechanism, especially the role of offering prices to the traders. Consider the following payoff specification:

- If traders  $(s, b)$  trade at price  $p$ ,
  - the seller gets  $p - s$ ;
  - the buyer gets  $b - p$ ;
  - the intermediary gets  $\lambda(p - s) + (1 - \lambda)(b - p)$ ;
- If no trade occurs, all players get 0.

The weight  $\lambda \in [0, 1]$  represents the degree of the intermediary's bias toward the seller. A biased intermediary as in the main model can be captured by assuming  $\lambda \in \left(\frac{1}{2}, 1\right]$ , and  $\lambda = \frac{1}{2}$  corresponds to the unbiased case. With this specification, I can obtain qualitatively the same results, namely:

- An ex post efficient acceptable CE exists if and only if the prior probability  $\pi_B$  of high type buyer is small enough,  $\pi_B \leq \widehat{\pi_B}$ ;
- There exists a threshold  $\bar{\lambda}$  on the degree of bias at which the SB outcome changes its regime, and the SB outcomes are those identified in Proposition 2;
- The expected trade surplus decreases if the bias changes from  $\lambda \leq \bar{\lambda}$  to  $\lambda' > \bar{\lambda}$ .

The important difference is that the threshold  $\widehat{\pi_B}$  is independent of the bias  $\lambda$ , which implies ex post efficiency can be achieved even when the intermediary's bias is extreme, say  $\lambda = 1$ .

This specification is also suitable for comparison with some work in the literature such as Matsuo (1989) since the intermediary's payoff is a straightforward extension of it. I discuss the relationship in Appendix C.1.

### 5.2.2 Intermediary in international relations

Kydd (2003) studies a biased intermediary in the context of international relations and shows that an intermediary is effective in reducing the probability of conflict only when he is biased.

Motivated by his work, consider the following payoff specification:

- If traders  $(s, b)$  trade at price  $p$ ,
  - the seller gets  $p - s$ ;
  - the buyer gets  $b - p$ ;
  - the intermediary gets  $\lambda(p - s)$ ;
- If no trade occurs,
  - the seller and the buyer get 0;
  - the intermediary gets  $-c$ , where  $c \in \mathbb{R}_{++}$ .

The weight  $\lambda \in [0, 1]$  represents the degree of the intermediary's bias as before. The intermediary is biased toward the seller if  $\lambda > 0$  and unbiased if  $\lambda = 0$ . His preference is aligned with the seller, but he incurs cost  $c$  when there is no trade. Contrary to Kydd (2003), I can show that a biased intermediary is not effective in achieving ex post efficiency. The crucial difference is that traders do not need to know the opponent's type to achieve ex post efficiency, whereas in Kydd (2003) the intermediary needs to credibly communicate his private information to reduce the probability of conflict, which is the reason why bias is necessary.<sup>24</sup>

There are many other payoff specifications under which qualitatively the same results can be obtained. What is important for the intermediary's preference is (i) he prefers trade to no trade; and (ii) his payoff is increasing in price. Combined with no commitment, this gives him the incentive to offer either of the buyer's highest acceptable prices ( $\overline{p_{b_L}}$  or  $\overline{p_{b_H}}$  in the main model). Using this fact, I can obtain the thresholds for the prior probability or the parameter governing the intermediary's bias under which a particular mediation plan constitutes an acceptable CE.

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<sup>24</sup>It is also worth mentioning that Kydd (2003) focuses on a fixed communication protocol whereas the present paper considers all admissible communication protocols, leveraging the communication revelation principle.

## 6 Conclusion

Incorporating a biased intermediary into a canonical model of bilateral trade, I have shown two main results. First, the biased mediation makes it harder to achieve ex post efficiency. Second, I characterize the SB outcome and show that the expected social surplus is weakly decreasing in the intermediary's bias. However, having a biased intermediary is better than having no intermediary if the degree of bias is small enough. Being biased does not completely invalidate his facilitative role.

The purpose of this paper was to study the implication of having a biased intermediary without commitment and enforcement power on bargaining outcome. The overall implication is ambiguous. On the one hand, the results of the SB case imply that even a biased intermediary can be useful, justifying the widespread use of intermediaries when their biases are common knowledge, as in the case of real estate agents. On the other hand, when it comes to achieving ex post efficiency, the traders can be better off without a biased intermediary. One way to reconcile this conflict is to incorporate the search cost into the case of no intermediary. If the traders do not rely on the intermediary, they need to find counterparties before they engage in any bargaining, which is costly to them. Intermediaries usually know markets better than traders and can help them in this respect. If the search cost for the intermediary, which is reflected in the commissions, is sufficiently smaller than the traders', having a biased intermediary is better than having no intermediary in the first-best case too. Also, on top of offering a price, intermediaries may be able to provide some information not available to traders based on their past experiences. If there is such private information, the traders may find it beneficial to employ intermediaries even when they are known to be biased. Relatedly, the comparison between a biased and an unbiased intermediary lacks an unambiguous conclusion. Depending on the commission structures, a biased intermediary can perform better than an unbiased one or not. Endogenizing the commission structures, which is consistent with the new rule for Realtors in the U.S. (see footnote 2), may help understand the comparison better. I plan to explore these extensions in future work.

# Appendix

## A Proofs

### A.1 Proof of Lemma 1

I show the contraposition. Consider an arbitrary mediation plan  $\mu \in \Delta(Q \times \{r^A\})$  such that there exists some  $q \in \text{supp}(\mu)$  and some  $(s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\}$  satisfying  $q(s, b) = p \notin \{\overline{p_{b_L}}, \overline{p_{b_H}}\}$ , where  $\text{supp}(\mu)$  is the support of  $\mu$ . I show that the intermediary can gain by disobeying the recommendation if he is recommended  $p$ . When he is recommended  $p$ , his posterior belief that the traders are  $(\tilde{s}, \tilde{b})$  is given by

$$\xi(\tilde{s}, \tilde{b} \mid p) = \frac{\Pr(\tilde{s}, \tilde{b}) \sum_{q \in \{q' \in Q: q'(\tilde{s}, \tilde{b})=p\}} \mu(q)}{\sum_{(s, b) \in \Theta_S \times \Theta_B} \Pr(s, b) \sum_{q \in \{q' \in Q: q'(s, b)=p\}} \mu(q)},$$

where  $\mu(q)$  denotes the probability that the mediator picks up pure mediation plan  $(q, r^A)$  when she uses  $\mu$ . Note that by assumption at least one of  $\xi(s_H, b_H \mid p)$ ,  $\xi(s_L, b_H \mid p)$ , and  $\xi(s_L, b_L \mid p)$  is positive. Given these beliefs, if the traders' are honest and obedient to the mediator, the intermediary's expected payoff from offering  $\tilde{p}$  is

$$\begin{cases} [\xi(s_H, b_H \mid p) + \xi(s_L, b_H \mid p)] (\delta_S + \delta_B) \tilde{p} & \text{if } \tilde{p} \in [\underline{p_{s_H}}, \overline{p_{b_H}}] \\ \xi(s_L, b_H \mid p) (\delta_S + \delta_B) \tilde{p} & \text{if } \tilde{p} \in (\overline{p_{b_L}}, \underline{p_{s_H}}) \\ [\xi(s_L, b_H \mid p) + \xi(s_L, b_L \mid p)] (\delta_S + \delta_B) \tilde{p} & \text{if } \tilde{p} \in [\underline{p_{s_L}}, \overline{p_{b_L}}] \\ 0 & \text{if } \tilde{p} \notin [\underline{p_{s_L}}, \overline{p_{b_H}}]. \end{cases}$$

From these payoffs, it is easy to see:

- if  $p \notin [\underline{p_{s_L}}, \overline{p_{b_H}}]$ , he can gain by offering some  $\tilde{p} \in [\underline{p_{s_L}}, \overline{p_{b_H}}]$ ;
- if  $p \in (\overline{p_{b_L}}, \underline{p_{s_H}})$ , he can gain by offering either  $\overline{p_{b_L}}$  or  $\overline{p_{b_H}}$ ;
- if  $p \in [\underline{p_{s_H}}, \overline{p_{b_H}})$  or  $p \in [\underline{p_{s_L}}, \overline{p_{b_L}})$ , he can gain by offering either  $\overline{p_{b_L}}$  or  $\overline{p_{b_H}}$ .

Thus  $\mu$  is not a CE.

## A.2 Proof of Lemma 2

Consider an arbitrary mediation plan  $\mu \in \Delta(Q \times \{r^A\})$  that satisfies (3.1) for all  $q \in \text{supp}(\mu)$ . By construction, all players get the same expected payoffs in  $\mu$  and  $\mu_x$  if they are honest and obedient to the mediator. Hence, it suffices to show that  $\mu$  has weakly more room for profitable manipulations for all players than  $\mu_x$ . Let  $x_{HH} \equiv x_{HH}(\mu) = x_{HH}(\mu_x)$ ,  $x_{LH} \equiv x_{LH}(\mu) = x_{LH}(\mu_x)$ , and  $x_{LL} \equiv x_{LL}(\mu) = x_{LL}(\mu_x)$ . The IC constraints for the high-type seller, the low-type buyer, and the intermediary are the same in  $\mu$  and  $\mu_x$ . I show that the high-type buyer and the low-type seller have weakly more room for profitable manipulations, hence weakly more stringent IC constraints in  $\mu$  than in  $\mu_x$ .

Consider the high-type buyer  $b_H$ . Suppose he misreports his type in  $\mu_x$ . Since  $x_q(s_H, b_L) \in (\overline{p_{b_H}}, +\infty)$  for all  $x_q \in \text{supp}(\mu_x)$ , he can secure a positive payoff of  $b_H - (1 + \delta_B)\overline{p_{b_L}} = b_H - b_L$  only when the seller is of low type and price  $\overline{p_{b_L}}$  is offered. Hence, his expected payoff from misreporting is at most

$$(1 - \pi_S)(1 - x_{LL})(b_H - b_L).$$

In  $\mu$ , however, he may be able to get a positive payoff when the seller is of high type; he can secure  $b_H - (1 + \delta_B)p$  if  $p \in [\underline{p_{s_H}}, \overline{p_{b_H}})$  is offered. Hence, his expected payoff from misreporting in  $\mu$  is at most

$$(1 - \pi_S)(1 - x_{LL})(b_H - b_L) + \pi_S \sum_{p \in [\underline{p_{s_H}}, \overline{p_{b_H}})} \sum_{q \in \{q' \in Q: q'(s_H, b_L) = p\}} \mu(q)[b_H - (1 + \delta_B)p],$$

where the second term is nonnegative. Thus, high-type buyer's IC constraint in  $\mu$  is at least as stringent as that in  $\mu_x$ .

Next, consider the low-type seller  $s_L$ . Suppose he misreports his type in  $\mu_x$ . Since  $x_q(s_H, b_L) \in (\overline{p_{b_H}}, +\infty)$  for all  $x_q \in \text{supp}(\mu_x)$ , he can secure a positive payoff only when the buyer is of high type; he can get  $(1 - \delta_S)\overline{p_{b_H}} - s_L = \frac{b_H}{h} - s_L$  if price  $\overline{p_{b_H}}$  is offered and  $(1 - \delta_S)\overline{p_{b_L}} - s_L = \frac{b_L}{h} - s_L$  if  $\overline{p_{b_L}}$  is offered. Hence, his expected payoff from misreporting is at most

$$\frac{\pi_B}{h} [x_{HH}b_H + (1 - x_{HH})b_L - hs_L].$$

In  $\mu$ , however, he may be able to get a positive payoff when the buyer is of low type; he can secure  $(1 - \delta_S)p - s_L$  if  $p \in (\underline{p_{s_L}}, \overline{p_{b_L}}]$  is offered. Hence, his expected payoff from misreporting in  $\mu$  is at most

$$\begin{aligned} & \frac{\pi_B}{h} [x_{HH}b_H + (1 - x_{HH})b_L - hs_L] \\ & + (1 - \pi_B) \sum_{p \in (\underline{p_{s_L}}, \overline{p_{b_L}}]} \sum_{q \in \{q' \in Q: q'(s_H, b_L) = p\}} \mu(q) [(1 - \delta_S)p - s_L], \end{aligned}$$

where the second term is nonnegative. Thus, low-type seller's IC constraint in  $\mu$  is at least as stringent as that in  $\mu_x$ .

The above argument suffices for the proof of Lemma 2. However, I spell out the players' IC constraints in  $\mu_x$  in order for the coming analysis.

**High-type seller  $s_H$**  If he is honest and obedient to the mediator, he gets  $(1 - \delta_S)\overline{p_{b_H}} - s_H$  when the buyer is of high type and price  $\overline{p_{b_H}}$  is offered and gets 0 otherwise. If he misreports his type, he can secure at most  $(1 - \delta_S)\overline{p_{b_H}} - s_H$  if the buyer is of high type and  $\overline{p_{b_H}}$  is offered. Hence, he has no incentive to manipulate  $\mu_x$  if

$$\pi_B x_{HH} [(1 - \delta_S)\overline{p_{b_H}} - s_H] \geq \pi_B x_{LH} [(1 - \delta_S)\overline{p_{b_H}} - s_H] \iff x_{HH} \geq x_{LH}. \quad (\text{IC-}s_H)$$

**High-type buyer  $b_H$**  If he is honest and obedient to the mediator, he gets  $b_H - b_L$  if the seller is of low type and  $\overline{p_{b_L}}$  is offered and gets 0 otherwise. Combined with the preceding argument, he has no incentive to manipulate  $\mu_x$  if

$$(1 - \pi_S)(1 - x_{LH})(b_H - b_L) \geq (1 - \pi_S)(1 - x_{LL})(b_H - b_L) \iff x_{LL} \geq x_{LH}. \quad (\text{IC-}b_H)$$

**Low-type seller  $s_L$**  If he is honest and obedient to the mediator, he gets  $\frac{b_H}{h} - s_L$  when the buyer is of high type and  $\overline{p_{b_H}}$  is offered, gets  $\frac{b_L}{h} - s_L$  when  $\overline{p_{b_L}}$  is offered regardless of the buyer's type, and gets 0 otherwise. Combined with the preceding argument, he has no incentive to manipulate

$\mu_x$  if

$$\begin{aligned}
& \pi_B x_{LH} (b_H - h_{SL}) + [\pi_B (1 - x_{LH}) + (1 - \pi_B) (1 - x_{LL})] (b_L - h_{SL}) \\
& \geq \pi_B [x_{HH} b_H + (1 - x_{HH}) b_L - h_{SL}] \\
& \iff (1 - \pi_B) (b_L - h_{SL}) (1 - x_{LL}) \geq \pi_B (b_H - b_L) (x_{HH} - x_{LH}). \quad (\text{IC-}s_L)
\end{aligned}$$

**Low-type buyer  $b_L$**  His expected payoff is 0 if he is honest and obedient to the mediator, and his expected payoff is at most 0 whatever manipulation he uses. Hence, his IC constraint is trivially satisfied.

**Intermediary** His posterior beliefs when he is recommended  $\overline{p_{b_H}}$  are given by

$$\begin{aligned}
\xi(s_H, b_H | b_H) &= \frac{\pi_S \pi_B x_{HH}}{\pi_S \pi_B x_{HH} + (1 - \pi_S) \pi_B x_{LH} + (1 - \pi_S) (1 - \pi_B) x_{LL}} \\
\xi(s_L, b_H | b_H) &= \frac{(1 - \pi_S) \pi_B x_{LH}}{\pi_S \pi_B x_{HH} + (1 - \pi_S) \pi_B x_{LH} + (1 - \pi_S) (1 - \pi_B) x_{LL}} \\
\xi(s_L, b_L | b_H) &= \frac{(1 - \pi_S) (1 - \pi_B) x_{LL}}{\pi_S \pi_B x_{HH} + (1 - \pi_S) \pi_B x_{LH} + (1 - \pi_S) (1 - \pi_B) x_{LL}} \\
\xi(s_H, b_L | b_H) &= 0.
\end{aligned}$$

By a similar argument as in the proof of Lemma 1, the intermediary finds it optimal to offer either  $\overline{p_{b_L}}$  or  $\overline{p_{b_H}}$ . Hence, following the recommendation is optimal for him if he prefers offering  $\overline{p_{b_H}}$  to  $\overline{p_{b_L}}$ :

$$\begin{aligned}
& [\xi(s_H, b_H | b_H) + \xi(s_L, b_H | b_H)] (\delta_S + \delta_B) \overline{p_{b_H}} \\
& \geq [\xi(s_L, b_H | b_H) + \xi(s_L, b_L | b_H)] (\delta_S + \delta_B) \overline{p_{b_L}} \\
& \iff [\pi_S \pi_B x_{HH} + (1 - \pi_S) \pi_B x_{LH}] b_H \\
& \geq [(1 - \pi_S) \pi_B x_{LH} + (1 - \pi_S) (1 - \pi_B) x_{LL}] b_L \\
& \iff \pi_S \pi_B b_H x_{HH} + (1 - \pi_S) \pi_B (b_H - b_L) x_{LH} \geq (1 - \pi_S) (1 - \pi_B) b_L x_{LL}. \quad (\text{IC-Int1})
\end{aligned}$$

Similarly, his posterior beliefs when he is recommended  $\overline{p_{b_L}}$  are given by

$$\begin{aligned}\xi(s_H, b_H | b_L) &= \frac{\pi_S \pi_B (1 - x_{HH})}{\pi_S \pi_B (1 - x_{HH}) + (1 - \pi_S) \pi_B (1 - x_{LH}) + (1 - \pi_S)(1 - \pi_B)(1 - x_{LL})} \\ \xi(s_L, b_H | b_L) &= \frac{(1 - \pi_S) \pi_B (1 - x_{LH})}{\pi_S \pi_B (1 - x_{HH}) + (1 - \pi_S) \pi_B (1 - x_{LH}) + (1 - \pi_S)(1 - \pi_B)(1 - x_{LL})} \\ \xi(s_L, b_L | b_L) &= \frac{(1 - \pi_S)(1 - \pi_B)(1 - x_{LL})}{\pi_S \pi_B (1 - x_{HH}) + (1 - \pi_S) \pi_B (1 - x_{LH}) + (1 - \pi_S)(1 - \pi_B)(1 - x_{LL})} \\ \xi(s_H, b_L | b_L) &= 0.\end{aligned}$$

By the same argument as in the previous case, following the recommendation is optimal for him if he prefers offering  $\overline{p_{b_L}}$  to  $\overline{p_{b_H}}$ :

$$\begin{aligned}& [\xi(s_L, b_H | b_L) + \xi(s_L, b_L | b_L)] (\delta_S + \delta_B) \overline{p_{b_L}} \\ & \geq [\xi(s_H, b_H | b_L) + \xi(s_L, b_H | b_L)] (\delta_S + \delta_B) \overline{p_{b_H}} \\ \iff & [(1 - \pi_S) \pi_B (1 - x_{LH}) + (1 - \pi_S)(1 - \pi_B)(1 - x_{LL})] b_L \\ & \geq [\pi_S \pi_B (1 - x_{HH}) + (1 - \pi_S) \pi_B (1 - x_{LH})] b_H \\ \iff & (1 - \pi_S)(1 - \pi_B) b_L (1 - x_{LL}) \geq \pi_S \pi_B b_H (1 - x_{HH}) + (1 - \pi_S) \pi_B (b_H - b_L)(1 - x_{LH}).\end{aligned}\tag{IC-Int2}$$

When he is recommended  $p \notin \{\overline{p_{b_L}}, \overline{p_{b_H}}\}$ , he believes that the traders are  $(s_L, b_H)$  with probability 1. Since they accept no offer, following the recommendation is always optimal for him.

### A.3 Proof of Proposition 1

If  $(x_{HH}, x_{LH}, x_{LL}) = (1, 0, 0)$ , then (IC- $s_H$ ), (IC- $b_H$ ), and (IC-Int1) are clearly satisfied. (IC- $s_L$ ) is reduced to

$$(1 - \pi_B)(b_L - h_{s_L}) \geq \pi_B(b_H - b_L) \iff \pi_B \leq \frac{b_L - h_{s_L}}{b_H - h_{s_L}}. \tag{A.1}$$

(IC-Int2) is reduced to

$$(1 - \pi_S)(1 - \pi_B)b_L \geq (1 - \pi_S)\pi_B(b_H - b_L) \iff \pi_B \leq \frac{b_L}{b_H}. \quad (\text{A.2})$$

Since  $\frac{b_L - h_{SL}}{b_H - h_{SL}} < \frac{b_L}{b_H}$ , an ex post efficient acceptable CE exists if and only if (A.1) holds.

#### A.4 Proof of Proposition 2

First, I characterize the solution to the linear program (P) when  $\pi_B > \overline{\pi_B}$ . Rearranging the constraints, (P) can be written as follows:

$$\begin{aligned} \max_{x = (x_{HH}, x_{LH}, x_{LL}) \in \mathbb{R}^3} \quad & \pi_S \pi_B (b_H - s_H) x_{HH} + (1 - \pi_S) \pi_B (b_H - s_L) \\ & + (1 - \pi_S)(1 - \pi_B)(b_L - s_L)(1 - x_{LL}) \\ \text{subject to} \quad & g_1(x) = x_{HH} - x_{LH} \geq 0 \\ & g_2(x) = x_{LL} - x_{LH} \geq 0 \\ & g_3(x) = (1 - \pi_B)(b_L - h_{SL})(1 - x_{LL}) \\ & \quad - \pi_B(b_H - b_L)(x_{HH} - x_{LH}) \geq 0 \\ & g_4(x) = \pi_S \pi_B b_H x_{HH} + (1 - \pi_S) \pi_B (b_H - b_L) x_{LH} - (1 - \pi_S)(1 - \pi_B) b_L x_{LL} \geq 0 \\ & g_5(x) = -\pi_S \pi_B b_H (1 - x_{HH}) - (1 - \pi_S) \pi_B (b_H - b_L)(1 - x_{LH}) \\ & \quad + (1 - \pi_S)(1 - \pi_B) b_L (1 - x_{LL}) \geq 0 \\ & g_6(x) = 1 - x_{HH} \geq 0 \\ & g_7(x) = 1 - x_{LH} \geq 0 \\ & g_8(x) = 1 - x_{LL} \geq 0 \\ & g_9(x) = x_{HH} \geq 0 \\ & g_{10}(x) = x_{LH} \geq 0 \\ & g_{11}(x) = x_{LL} \geq 0 \end{aligned}$$

Let  $\eta_i$  denote the Lagrange multiplier associated with the  $g_i$ . By the KKT conditions,  $(x, \eta) \in \mathbb{R}^3 \times \mathbb{R}_+^{11}$  is the solution to the program if and only if it satisfies

**Primal feasibility:**  $g_i(x) \geq 0$  for all  $i$ .

**Dual feasibility:**  $\eta_i \geq 0$  for all  $i$ , and

$$\pi_S \pi_B (b_H - s_H) + \eta_1 - \pi_B (b_H - b_L) \eta_3 + \pi_S \pi_B b_H (\eta_4 + \eta_5) - \eta_6 + \eta_9 = 0 \quad (\text{Dual 1})$$

$$- \eta_1 - \eta_2 + \pi_B (b_H - b_L) \eta_3 + (1 - \pi_S) \pi_B (b_H - b_L) (\eta_4 + \eta_5) - \eta_7 + \eta_{10} = 0 \quad (\text{Dual 2})$$

$$\begin{aligned} & - (1 - \pi_S) (1 - \pi_B) (b_L - s_L) + \eta_2 - (1 - \pi_B) (b_L - h s_L) \eta_3 \\ & - (1 - \pi_S) (1 - \pi_B) b_L (\eta_4 + \eta_5) - \eta_8 + \eta_{11} = 0. \end{aligned} \quad (\text{Dual 3})$$

**Complementary slackness:**  $\eta_i g_i(x) = 0$  for all  $i$ .

For the sake of exposition, I introduce the following notations:

$$\overline{x_{HH}} = \frac{(1 - \pi_B) (b_L - h s_L)}{\pi_B (b_H - b_L)}$$

$$J = \pi_S \pi_B b_H + (1 - \pi_S) \pi_B (b_H - b_L) - (1 - \pi_S) (1 - \pi_B) b_L$$

$$K = \pi_S \pi_B s_H (1 - \overline{x_{HH}}) + (1 - \pi_S) \pi_B (b_H - b_L) - (1 - \pi_S) (1 - \pi_B) s_L.$$

Note that  $\overline{x_{HH}}$  is the value of  $x_{HH}$  at which  $g_3(x) \geq 0$  binds when  $x_{LH} = x_{LL} = 0$  and that  $-J$  is the sum of the constant terms in  $g_5(x)$ . As such, the larger the value of  $J$ , the larger the intermediary's incentive to disobey when  $\overline{p_{b_L}}$  is recommended.

**Case 1: When  $J \leq \pi_S \pi_B b_H \overline{x_{HH}} + \min\{K, 0\}$**

I show that

$$x^* = (\overline{x_{HH}}, 0, 0)$$

$$\eta_2^* = \pi_S \pi_B (b_H - s_H)$$

$$\eta_3^* = \frac{\pi_S (b_H - s_H)}{b_H - b_L}$$

$$\eta_{11}^* = (1 - \pi_S) (1 - \pi_B) (b_L - s_L) - \pi_S \pi_B (b_H - s_H) (1 - \overline{x_{HH}})$$

$$\eta_i^* = 0 \text{ for all } i \notin \{2, 3, 11\}$$

are the solution to the program.

**Primal feasibility:** It is easy to see  $g_i(x^*) \geq 0$  for all  $i \neq \{5, 6\}$ . Note that  $g_5(x^*) \geq 0$  is reduced to  $J \leq \pi_S \pi_B b_H \overline{x_{HH}}$  and that  $g_6(x^*) = 1 - \overline{x_{HH}} \geq 0 \iff \pi_B \geq \overline{\pi_B}$ .

**Dual feasibility:**  $\eta_2^*$  and  $\eta_3^*$  are positive, and

$$\eta_{11} \geq 0 \iff J \leq \pi_S \pi_B b_H \overline{x_{HH}} + K.$$

The left-side of (Dual 1) is reduced to

$$\pi_S \pi_B (b_H - s_H) - \pi_S \pi_B (b_H - s_H) = 0.$$

(Dual 2) is satisfied since  $\eta_2^* = \pi_B (b_H - b_L) \eta_3^*$ . The left-side of (Dual 3) is reduced to

$$\pi_S \pi_B (b_H - s_H) \overline{x_{HH}} - \underbrace{(1 - \pi_B)(b_L - s_L)}_{= \pi_B (b_H - b_L) \overline{x_{HH}}} \frac{\pi_S (b_H - s_H)}{b_H - b_L} = 0.$$

**Complementary slackness:** The conditions are satisfied since  $\eta_i^* = 0$  for all  $i \notin \{2, 3, 11\}$  and  $g_2(x^*) = g_3(x^*) = g_{11}(x^*) = 0$ .

**Case 2: When  $\pi_S \pi_B b_H \overline{x_{HH}} < J \leq \pi_S \pi_B b_H \overline{x_{HH}} + K$**

I show that

$$\begin{aligned} x^* &= (1, 1, 1) \\ \eta_2^* &= \pi_S \pi_B (b_H - s_H) + [\pi_S \pi_B b_H + (1 - \pi_S) \pi_B (b_H - b_L)] \eta_5^* \\ \eta_3^* &= \frac{\pi_S \pi_B (b_H - s_H) + \pi_S \pi_B b_H \eta_5^*}{\pi_B (b_H - b_L)} \\ \eta_5^* &= \frac{(1 - \pi_S)(1 - \pi_B)(b_L - s_L) - \pi_S \pi_B (b_H - s_H)(1 - \overline{x_{HH}})}{J - \pi_S \pi_B b_H \overline{x_{HH}}} \\ \eta_i^* &= 0 \text{ for all } i \notin \{2, 3, 5\} \end{aligned}$$

are the solution to the program.

**Primal feasibility:** It is easy to see  $g_i(x^*) \geq 0$  for all  $i \neq 4$  and  $g_4(x^*) = J > 0$  by assumption.

**Dual feasibility:** Note that  $\eta_2^*$  and  $\eta_3^*$  are positive if  $\eta_5^*$  is nonnegative, which is the case if

$$\pi_S \pi_B b_H \overline{x_{HH}} < J \leq \pi_S \pi_B b_H \overline{x_{HH}} + K.$$

The left-side of (Dual 1) is reduced to

$$\pi_S \pi_B (b_H - s_H) - \pi_S \pi_B (b_H - s_H) - \pi_S \pi_B b_H \eta_5^* + \pi_S \pi_B b_H \eta_5^* = 0.$$

The left-side of (Dual 2) is reduced to

$$\begin{aligned} & -\pi_S \pi_B (b_H - s_H) - [\pi_S \pi_B b_H + (1 - \pi_S) \pi_B (b_H - b_L)] \eta_5^* \\ & + \pi_S \pi_B (b_H - s_H) + \pi_S \pi_B b_H \eta_5^* + (1 - \pi_S) \pi_B (b_H - b_L) \eta_5^* = 0. \end{aligned}$$

The left-side of (Dual 3) is reduced to

$$\begin{aligned} & -(1 - \pi_S)(1 - \pi_B)(b_L - s_L) + \pi_S \pi_B (b_H - s_H) + [\pi_S \pi_B b_H + (1 - \pi_S) \pi_B (b_H - b_L)] \eta_5^* \\ & - \underbrace{(1 - \pi_B)(b_L - s_L)}_{= \pi_B (b_H - b_L) \overline{x_{HH}}} \frac{\pi_S \pi_B (b_H - s_H) + \pi_S \pi_B b_H \eta_5^*}{\pi_B (b_H - b_L)} - (1 - \pi_S)(1 - \pi_B) b_L \eta_5^* \\ & = \pi_S \pi_B (b_H - s_H)(1 - \overline{x_{HH}}) - (1 - \pi_S)(1 - \pi_B)(b_L - s_L) + [J - \pi_S \pi_B b_H \overline{x_{HH}}] \eta_5^* \\ & = 0. \end{aligned}$$

**Complementary slackness:** The conditions are satisfied since  $\eta_i^* = 0$  for all  $i \notin \{2, 3, 5\}$  and

$$g_2(x^*) = g_3(x^*) = g_5(x^*) = 0.$$

**Case 3: When  $J > \pi_S \pi_B b_H \overline{x_{HH}} + K$**

I show that

$$x^* = (1, 1, 1)$$

$$\eta_2^* = \pi_S \pi_B (b_H - s_H)$$

$$\eta_3^* = \frac{\pi_S (b_H - s_H)}{b_H - b_L}$$

$$\eta_8^* = \pi_S \pi_B (b_H - s_H) (1 - \overline{x_{HH}}) - (1 - \pi_S) (1 - \pi_B) (b_L - s_L)$$

$$\eta_i^* = 0 \text{ for all } i \notin \{2, 3, 8\}$$

are the solution to the program.

**Primal feasibility:** It is easy to see  $g_i(x^*) \geq 0$  for all  $i \neq 4$ . Since  $K < 0$  implies  $\pi_S \pi_B b_H \overline{x_{HH}} + K > 0$  and  $g_4(x^*) = J > 0$  regardless of the sign of  $K$ .

**Dual feasibility:** Note that  $\eta_2^*$  and  $\eta_3^*$  coincide with those in Case 1 and that  $-\eta_8^*$  is equal to “ $\eta_{11}^*$  in Case 1.” Thus, the dual feasibility is satisfied if

$$J \geq \pi_S \pi_B b_H \overline{x_{HH}} + K.$$

**Complementary slackness:** The conditions are satisfied since  $\eta_i^* = 0$  for all  $i \notin \{2, 3, 8\}$  and  $g_2(x^*) = g_3(x^*) = g_8(x^*) = 0$ .

In summary, the solution to the linear program (P) when  $\pi_B > \overline{\pi_B}$  is

- $(x_{HH}^*, x_{LH}^*, x_{LL}^*) = (\overline{x_{HH}}, 0, 0)$  if  $J \leq \pi_S \pi_B b_H \overline{x_{HH}} + \min\{K, 0\}$ ;
- $(x_{HH}^*, x_{LH}^*, x_{LL}^*) = (1, 1, 1)$  otherwise.

In the former case, traders  $(s_H, b_H)$  do not trade with probability  $1 - \overline{x_{HH}}$  while  $(s_L, b_H)$  and  $(s_L, b_L)$  trade at price  $\overline{p_{b_L}}$  with probability 1. In the latter case, traders  $(s_H, b_H)$  and  $(s_L, b_H)$  trade at price  $\overline{p_{b_H}}$  with probability 1 while  $(s_L, b_L)$  do not trade. Note that  $J \leq \pi_S \pi_B b_H \overline{x_{HH}} +$

$\min\{K, 0\}$  implies two upper bounds on the ratio  $h = \frac{1+\delta_B}{1-\delta_S}$ :

$$J \leq \pi_S \pi_B b_H \overline{x_{HH}} \iff h \leq \frac{b_L}{s_L} - \frac{(b_H - b_L)J}{\pi_S(1 - \pi_B)b_H s_L} \equiv h_1 \quad (\text{A.3})$$

and

$$\begin{aligned} J &\leq \pi_S \pi_B b_H \overline{x_{HH}} + K \\ \iff h &\leq \frac{b_L}{s_L} - \frac{(b_H - b_L) [\pi_S \pi_B (b_H - s_H) - (1 - \pi_S)(1 - \pi_B)(b_L - s_L)]}{\pi_S(1 - \pi_B)(b_H - s_H)s_L} \equiv h_2. \end{aligned} \quad (\text{A.4})$$

Thus, the threshold is given by  $\bar{h} \equiv \min\{h_1, h_2\}$ .

## A.5 Proof of Corollary 1

For  $\left(\bar{h}, \min\left\{\frac{b_L}{s_L}, \frac{b_H}{s_H}\right\}\right]$ , the expected social surplus is given by

$$\pi_S \pi_B (b_H - s_H) + (1 - \pi_S) \pi_B (b_H - s_L),$$

which is independent of  $h$ . For  $h \leq \bar{h}$ , the expected social surplus is given by

$$\pi_S \pi_B (b_H - s_H) \overline{x_{HH}} + (1 - \pi_S) \pi_B (b_H - s_L) + (1 - \pi_S)(1 - \pi_B)(b_L - s_L).$$

Since  $\overline{x_{HH}}$  is decreasing in  $h$  on  $(1, \bar{h})$ , so is the expected social surplus. Finally, if  $h \leq \bar{h}$  increases to  $h' > \bar{h}$ , the expected social surplus changes by

$$\pi_S \pi_B (b_H - s_H)(1 - \overline{x_{HH}}) - (1 - \pi_S)(1 - \pi_B)(b_L - s_L),$$

which is nonpositive if  $J \leq \pi_S \pi_B b_H \overline{x_{HH}} + K$ . Thus, the expected social surplus decreases unless  $h = \bar{h} = h_2$ .

## B On CE

In Section B.1, I show that any CE is SCE in the mediated bargaining game. In Section B.2, I show the equivalence between ex ante IC constraints and the interim IC constraints.

### B.1 Equivalence between CE and SCE

Myerson (1986b, Theorem 2) shows that a CE is a SCE if and only if it never recommends a “codominated action” to any player who has not lied to the mediator. Thus, it suffices to show that no action is codominated in the mediated bargaining game.

To define the concept of codomination, let  $C$  be a correspondence such that

$$C(h^0) \subseteq \mathbb{R}_+,$$

$$C(\theta_i, p, p') \subseteq \{Y, N\}, \forall i \in \{S, B\}, \forall (\theta_i, p, p') \in \Theta_i \times \mathbb{R}_+^2,$$

where  $h^0$  denotes the initial history. For any such  $C$ , let  $E(C)$  be the set of pure mediation plans that never recommend actions in  $C(\theta_i, p, p')$  for all  $i \in \{S, B\}$  and all  $(\theta_i, p, p') \in \Theta_i \times \mathbb{R}_+^2$ ; that is,

$$E(C) = \{(q, r) \in Q \times R : \forall (s, b, p, p') \in \Theta_S \times \Theta_B \times \mathbb{R}_+^2,$$

$$r_S(p' \mid s, b, p) \notin C(s, p, p'), r_B(p' \mid s, b, p) \notin C(b, p, p')\}.$$

For any  $p \in \mathbb{R}_+$ , let  $\phi^1(p)$  be the set of tuples consisting of a pure mediation plan and the traders' reports such that the mediator recommends  $p$ ; that is,

$$\phi^1(p) = \{(q, r, s, b) \in Q \times R \times \Theta_S \times \Theta_B : q(s, b) = p\}.$$

For each  $i \in \{S, B\}$ , for any  $a_i \in \{Y, N\}$  and any  $(\theta_i, p, p') \in \Theta_i \times \mathbb{R}_+^2$ , let  $\phi^2(a_i, \theta_i, p, p')$  be the set of tuples consisting of a pure mediation plan, the traders' types, the recommended price, and the intermediary's offer such that  $i$ 's type is  $\theta_i$  and the mediator recommends  $a_i$  to  $i$  as a

response to  $p'$  if the traders do not lie; that is, for  $i = S$ ,

$$\phi^2(a_S, \theta_S, p, p') = \{(q, r, s, b, p, p') \in Q \times R \times \Theta_S \times \Theta_B \times \mathbb{R}_+^2 : s = \theta_S, q(s, b) = p, r_S(p' | s, b, p) = a_S\},$$

and for  $i = B$ ,

$$\phi^2(a_B, \theta_B, p, p') = \{(q, r, s, b, p, p') \in Q \times R \times \Theta_S \times \Theta_B \times \mathbb{R}_+^2 : b = \theta_B, q(s, b) = p, r_B(p' | s, b, p) = a_B\}.$$

Let  $\phi^1(C)$  be the union of all sets  $\phi^1(p)$  over all  $p \in C(h^0)$  and  $\phi^2(C)$  be the union of all sets  $\phi^2(a_i, \theta_i, p, p')$  over all  $i \in \{S, B\}$ ,  $(\theta_i, p, p') \in \Theta_i \times \mathbb{R}_+^2$ , and  $a_i \in C(\theta_i, p, p')$ ; that is,

$$\phi^1(C) = \{(q, r, s, b) \in Q \times R \times \Theta_S \times \Theta_B : q(s, b) \in C(h^0)\},$$

$$\phi^2(C) = \{(q, r, s, b, p, p') \in Q \times R \times \Theta_S \times \Theta_B \times \mathbb{R}_+^2 :$$

$$r_S(p' | s, b, p) \in C(s, p, p') \text{ or } r_B(p' | s, b, p) \in C(b, p, p')\}.$$

Now I can define codomination correspondence.  $C$  is a *codomination correspondence* if

1. for each  $\tau^1 \in \Delta(Q \times R \times \Theta_S \times \Theta_B)$ , if  $\tau^1(E(C) \times \Theta_S \times \Theta_B) = 1$  and  $\tau^1(\phi^1(C)) > 0$ , then there exists some  $p \in C(h^0)$  and some  $\alpha : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  such that

$$\begin{aligned} & \sum_{(q,r,s,b) \in \phi^1(p)} \tau^1(q, r, s, b) \cdot (\delta_S + \delta_B) \alpha(p) \cdot \mathbf{1}_{\{r(\alpha(p)|s,b,p)=(Y,Y)\}} \\ & > \sum_{(q,r,s,b) \in \phi^1(p)} \tau^1(q, r, s, b) \cdot (\delta_S + \delta_B) p \cdot \mathbf{1}_{\{r(p|s,b,p)=(Y,Y)\}} \end{aligned} \quad (\text{B.1})$$

and

2. for each  $\tau^2 \in \Delta(Q \times R \times \Theta_S \times \Theta_B \times \mathbb{R}_+^2)$ , if  $\tau^2(\phi^2(C)) > 0$ , then there exists some trader  $i \in \{S, B\}$ , some  $(\theta_i, p, p') \in \Theta_i \times \mathbb{R}_+^2$ , some  $a_i \in C(\theta_i, p, p')$ , and some  $\gamma_i :$

$\Theta_i \times \mathbb{R}_+ \times \{Y, N\} \rightarrow \{Y, N\}$  such that, if  $i = S$ ,

$$\begin{aligned} & \sum_{(q,r,s,b,p,p') \in \phi^2(a_S, \theta_S, p, p')} \tau^2(q, r, s, b, p, p') \cdot [(1 - \delta_S)p - s] \cdot \mathbf{1}_{\{\gamma_S(s, p', a_S) = Y \cap r_B(p' | s, b, p) = Y\}} \\ & > \sum_{(q,r,s,b,p,p') \in \phi^2(a_S, \theta_S, p, p')} \tau^2(q, r, s, b, p, p') \cdot [(1 - \delta_S)p - s] \cdot \mathbf{1}_{\{a_S = Y \cap r_B(p' | s, b, p) = Y\}} \end{aligned} \quad (\text{B.2})$$

or, if  $i = B$ ,

$$\begin{aligned} & \sum_{(q,r,s,b,p,p') \in \phi^2(a_B, \theta_B, p, p')} \tau^2(q, r, s, b, p, p') \cdot [b - (1 + \delta_B)p] \cdot \mathbf{1}_{\{\gamma_B(b, p', a_B) = Y \cap r_S(p' | s, b, p) = Y\}} \\ & > \sum_{(q,r,s,b,p,p') \in \phi^2(a_B, \theta_B, p, p')} \tau^2(q, r, s, b, p, p') \cdot [b - (1 + \delta_B)p] \cdot \mathbf{1}_{\{a_B = Y \cap r_S(p' | s, b, p) = Y\}}. \end{aligned} \quad (\text{B.3})$$

Let  $D$  denote the union of all codomination correspondences.<sup>25</sup> The intermediary's offer  $p \in \mathbb{R}_+$  is *codominated* if  $p \in D(h^0)$ . Trader  $i$ 's response  $a_i \in \{Y, N\}$  is *codominated* at the history  $(\theta_i, p, p') \in \Theta_i \times \mathbb{R}_+^2$  if  $a_i \in D(\theta_i, p, p')$ . Roughly speaking, the definition requires that if codominated actions could be recommended with positive probability, then at least one player could expect to gain by manipulation after being told to use a codominated action.

Lemma 4 shows that no action is codominated in the mediated bargaining game, which implies that every CE is a SCE.

**Lemma 4.** *No action is codominated in the mediated bargaining game.*

*Proof.* I show that the only codomination correspondence in the mediated bargaining game is  $C$  such that

$$C(h^0) = \emptyset \quad (\text{B.4})$$

$$C(\theta_i, p, p') = \emptyset, \forall i \in \{S, B\}, \forall (\theta_i, p, p') \in \Theta_i \times \mathbb{R}_+^2. \quad (\text{B.5})$$

First, I show (B.5). Note that  $C(\theta_i, p, p') \subseteq \{Y, N\}$  is either  $\emptyset$ ,  $\{Y\}$ ,  $\{N\}$ , or  $\{Y, N\}$  for any

<sup>25</sup>In general, this union may not be well defined if type spaces or action spaces are not finite. However, there is a unique codomination correspondence in the present model (see Lemma 4).

$i \in \{S, B\}$  and any  $(\theta_i, p, p') \in \Theta_i \times \mathbb{R}_+^2$ . Fix arbitrary  $(s, b, p) \in \Theta_S \times \Theta_B \times \mathbb{R}_+$ .

1. If  $C(s, p, p') = \{Y\}$  and  $C(b, p, p') = \{Y\}$  or  $\emptyset$ , take  $\tau^2 \in \Delta(Q \times R \times \Theta_S \times \Theta_B \times \mathbb{R}_+^2)$  such that  $\tau^2(q, r, s, b, p, p') = 1$  for  $q(s, b) = p$  and  $r(p' \mid s, b, p) = (Y, N)$ . The seller is recommended an action in  $C$ , but no manipulation is profitable for him, which violates (B.2).
2. If  $C(s, p, p') = \emptyset$  and  $C(b, p, p') = \{Y\}$ , take  $\tau^2 \in \Delta(Q \times R \times \Theta_S \times \Theta_B \times \mathbb{R}_+^2)$  such that  $\tau^2(q, r, s, b, p, p') = 1$  for  $q(s, b) = p$  and  $r(p' \mid s, b, p) = (N, Y)$ . The buyer is recommended an action in  $C$ , but no manipulation is profitable for him, which violates (B.3).
3. In all the other cases except for  $C(s, p, p') = C(b, p, p') = \emptyset$ , take  $\tau^2 \in \Delta(Q \times R \times \Theta_S \times \Theta_B \times \mathbb{R}_+^2)$  such that  $\tau^2(q, r, s, b, p, p') = 1$  for  $q(s, b) = p$  and  $r(p' \mid s, b, p) = (N, N)$ . At least one trader is recommended an action in  $C$ , but no manipulation is profitable for him, which violates (B.2) or (B.3).
4. If  $C(s, p, p') = \emptyset$  and  $C(b, p, p') = \emptyset$ , then the definition is trivially satisfied.

Since I have chosen  $(s, b, p, p')$  arbitrarily, a codomination correspondence must satisfy  $C(s, p, p') = \emptyset$  and  $C(b, p, p') = \emptyset$  for all  $(s, b, p, p') \in \Theta_S \times \Theta_B \times \mathbb{R}_+^2$ .

Next, I show (B.4). Suppose  $C(h^0) \neq \emptyset$ . For each  $p \in C(h^0)$ , take  $\tau^1 \in \Delta(Q \times R \times \Theta_S \times \Theta_B)$  such that  $\tau^1(q, r, s, b) = 1$  for  $q(s, b) = p$  and  $r(p' \mid s, b, p) = (N, N)$  for all  $p' \in \mathbb{R}_+$ . The intermediary is recommended an action in  $C$ , but no manipulation is profitable for him, which violates (B.1). Hence,  $C(h^0)$  must be empty as well. Therefore, the correspondence  $C$  satisfying (B.4) and (B.5) is the only codomination correspondence in the mediated bargaining game.  $\square$

## B.2 Equivalence between ex ante IC and interim IC

As I only consider pure mediation plans or mixed mediation plans in  $\Delta(Q \times \{r^A\})$  in the paper, hereafter I focus on mediation plans  $\mu \in \Delta(Q \times \{r\})$  for some arbitrary  $r \in R$ . I show that the ex ante IC constraints (2.1) and (2.2) are satisfied if and only if the interim IC constraints are satisfied.

For trader  $i \in \{S, B\}$ , let  $U_i(q, r \mid \theta_i)$  be type  $\theta_i$ 's interim expected payoff if the mediator uses a pure mediation plan  $(q, r)$  and all players are honest and obedient to the mediator:

$$U_S(q, r \mid s) = \sum_{b \in \Theta_B} \Pr(b) \cdot [(1 - \delta_S)q(s, b) - s] \cdot \mathbf{1}_{\{r(q(s, b) \mid s, b, q(s, b)) = (Y, Y)\}},$$

$$U_B(q, r \mid b) = \sum_{s \in \Theta_S} \Pr(s) \cdot [b - (1 + \delta_B)q(s, b)] \cdot \mathbf{1}_{\{r(q(s, b) \mid s, b, q(s, b)) = (Y, Y)\}},$$

where  $\Pr(\theta_i)$  is the prior probability that trader  $i$  is of type  $\theta_i$ . Suppose that the mediator uses a mediation plan  $\mu \in \Delta(Q \times \{r\})$ . If all players are honest and obedient to the mediator, then trader  $i$  of type  $\theta_i$  expects to get

$$\sum_{q \in Q} \mu(q) U_i(q, r \mid \theta_i).$$

Let  $V(\mu \mid p)$  be the intermediary's interim expected payoff if he receives the recommendation  $p$  and all players are honest and obedient to the mediator:

$$V(\mu \mid p) = \sum_{(s, b) \in \Theta_S \times \Theta_B} \xi(s, b \mid p) \cdot (\delta_S + \delta_B)p \cdot \mathbf{1}_{\{r(p \mid s, b, p) = (Y, Y)\}},$$

where  $\xi(s, b \mid p)$  is the intermediary's posterior belief that the traders are  $(s, b)$  when he receives  $p$ , which is given by

$$\xi(s, b \mid p) = \frac{\Pr(s, b) \sum_{q \in \{q' \in Q : q'(s, b) = p\}} \mu(q)}{\sum_{(\tilde{s}, \tilde{b}) \in \Theta_S \times \Theta_B} \Pr(\tilde{s}, \tilde{b}) \sum_{q \in \{q' \in Q : q'(\tilde{s}, \tilde{b}) = p\}} \mu(q)}.$$

For each recommendation  $p$ , the intermediary's interim manipulation is represented by the price  $\hat{\alpha} \in \mathbb{R}_+$  he actually offers. For trader  $i \in \{S, B\}$  of type  $\theta_i \in \Theta_i$ , his interim manipulation is represented by a pair  $(\hat{\beta}_i, \hat{\gamma}_i)$ , where  $\hat{\beta}_i \in \Theta_i$  is a manipulation in report and  $\hat{\gamma}_i : \{Y, N\} \times \mathbb{R}_+ \rightarrow \{Y, N\}$  is a manipulation in response. Let  $\hat{\Sigma}_i$  be the set of all such pairs  $(\hat{\beta}_i, \hat{\gamma}_i)$ . For any  $(q, r) \in Q \times R$  and  $(\hat{\beta}_i, \hat{\gamma}_i) \in \hat{\Sigma}_i$ , the interim expected payoff  $U_i((q, r) \circ (\hat{\beta}_i, \hat{\gamma}_i) \mid \theta_i)$  of trader  $i$  of type  $\theta_i$  when he manipulates  $(q, r)$  by  $(\hat{\beta}_i, \hat{\gamma}_i)$  is naturally defined. If the intermediary receives the recommendation  $p$  and manipulates  $\mu$  by  $\hat{\alpha} \in \mathbb{R}_+$ , then he expects to get

$$V(\mu \circ \hat{\alpha} \mid p) = \sum_{(s, b) \in \Theta_S \times \Theta_B} \xi(s, b \mid p) \cdot (\delta_S + \delta_B)\hat{\alpha} \cdot \mathbf{1}_{\{r(\hat{\alpha} \mid s, b, p) = (Y, Y)\}}.$$

The intermediary has no incentive to manipulate  $\mu \in \Delta(Q \times \{r\})$  after he receives a recommendation (at Time 5) if, for all  $\hat{\alpha} \in \mathbb{R}_+$  and all  $p \in \mathbb{R}_+$  such that  $p = q(s, b)$  for some  $q \in \text{supp}(\mu)$  and some  $(s, b) \in \Theta_S \times \Theta_B$ ,

$$V(\mu \mid p) \geq V(\mu \circ \hat{\alpha} \mid p). \quad (\text{B.6})$$

Trader  $i \in \{S, B\}$  of type  $\theta_i \in \Theta_i$  has no incentive to manipulate  $\mu \in \Delta(Q \times \{r\})$  after he learns his type (at Time 3) if, for all  $(\hat{\beta}_i, \hat{\gamma}_i) \in \hat{\Sigma}_i$ ,

$$\sum_{q \in Q} \mu(q) U_i(q, r \mid \theta_i) \geq \sum_{q \in Q} \mu(q) U_i((q, r) \circ (\hat{\beta}_i, \hat{\gamma}_i) \mid \theta_i). \quad (\text{B.7})$$

**Lemma 5.** *For any mediation plan  $\mu \in \Delta(Q \times \{r\})$  for some arbitrary  $r \in R$ , the ex ante IC constraints (2.1) and (2.2) are satisfied if and only if the interim IC constraints (B.6) and (B.7) are satisfied.*

*Proof.* Suppose that (B.6) is not satisfied for  $\mu \in \Delta(Q \times \{r\})$ ; that is, there exists some recommendation  $\hat{p} \in \mathbb{R}_+$  and the intermediary's manipulation  $\hat{\alpha} \in \mathbb{R}_+$  such that

$$\begin{aligned} & V(\mu \circ \hat{\alpha} \mid \hat{p}) > V(\mu \mid \hat{p}) \\ \iff & \sum_{(s,b) \in \Theta_S \times \Theta_B} \xi(s, b \mid \hat{p}) \cdot (\delta_S + \delta_B) \hat{\alpha} \cdot \mathbf{1}_{\{r(\hat{\alpha}|s,b,\hat{p})=(Y,Y)\}} \\ & > \sum_{(s,b) \in \Theta_S \times \Theta_B} \xi(s, b \mid \hat{p}) \cdot (\delta_S + \delta_B) \hat{p} \cdot \mathbf{1}_{\{r(\hat{p}|s,b,\hat{p})=(Y,Y)\}}. \end{aligned} \quad (\text{B.8})$$

Consider an ex ante manipulation  $\alpha \in \Sigma_I$  of the intermediary such that

$$\alpha(p) = \begin{cases} \hat{\alpha} & \text{if } p = \hat{p} \\ p & \text{if } p \neq \hat{p}. \end{cases}$$

In other words, the intermediary disobeys only when he receives  $\hat{p}$ . Let  $Q_\mu(\hat{p})$  and  $Q_\mu(s, b, \hat{p})$  be the set of price-recommendations in the support of  $\mu$  that recommends  $\hat{p} \in \mathbb{R}_+$  for some pairs

of types and for particular pair  $(s, b) \in \Theta_S \times \Theta_B$ , respectively; that is,

$$Q_\mu(\hat{p}) = \{q \in Q : \mu(q) > 0 \text{ and } q(s, b) = \hat{p} \text{ for some } (s, b) \in \Theta_S \times \Theta_B\},$$

$$Q_\mu(s, b, \hat{p}) = \{q \in Q : \mu(q) > 0 \text{ and } q(s, b) = \hat{p}\}.$$

For any  $q \in Q$ , let  $\Theta_q(\hat{p})$  be the set of pairs of types for which  $q$  recommends  $\hat{p}$ ; that is,

$$\Theta_q(\hat{p}) = \{(s, b) \in \Theta_S \times \Theta_B : q(s, b) = \hat{p}\}.$$

Then, I have  $\sum_{q \in Q} \mu(q) V(q \circ \alpha, r) = \sum_{q \in Q_\mu(\hat{p})} \mu(q) V(q \circ \alpha, r) + \sum_{q \notin Q_\mu(\hat{p})} \mu(q) V(q, r)$ . I can further decompose the first term in the right side as

$$\begin{aligned} & \sum_{q \in Q_\mu(\hat{p})} \mu(q) V(q \circ \alpha, r) \\ &= \sum_{q \in Q_\mu(\hat{p})} \mu(q) \left[ \sum_{(s, b) \in \Theta_q(\hat{p})} \Pr(s, b) \cdot (\delta_S + \delta_B) \hat{\alpha} \cdot \mathbf{1}_{\{r(\hat{\alpha}|s, b, \hat{p}) = (Y, Y)\}} \right. \\ & \quad \left. + \sum_{(s, b) \notin \Theta_q(\hat{p})} \Pr(s, b) \cdot (\delta_S + \delta_B) q(s, b) \cdot \mathbf{1}_{\{r(q(s, b)|s, b, q(s, b)) = (Y, Y)\}} \right] \\ &= \sum_{(s, b) \in \cup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \Pr(s, b) \left[ \sum_{q \in Q_\mu(s, b, \hat{p})} \mu(q) \cdot (\delta_S + \delta_B) \hat{\alpha} \cdot \mathbf{1}_{\{r(\hat{\alpha}|s, b, \hat{p}) = (Y, Y)\}} \right. \\ & \quad \left. + \sum_{q \in Q_\mu(\hat{p}) \setminus Q_\mu(s, b, \hat{p})} \mu(q) \cdot (\delta_S + \delta_B) q(s, b) \cdot \mathbf{1}_{\{r(q(s, b)|s, b, q(s, b)) = (Y, Y)\}} \right] \\ & \quad + \sum_{(s, b) \notin \cup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \Pr(s, b) \sum_{q \in Q_\mu(\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) q(s, b) \cdot \mathbf{1}_{\{r(q(s, b)|s, b, q(s, b)) = (Y, Y)\}}. \end{aligned} \tag{B.9}$$

Since  $\xi(s, b | \hat{p})$  is given by

$$\xi(s, b | \hat{p}) = \frac{\Pr(s, b) \sum_{q \in Q_\mu(s, b, \hat{p})} \mu(q)}{\sum_{(\tilde{s}, \tilde{b}) \in \Theta_S \times \Theta_B} \Pr(\tilde{s}, \tilde{b}) \sum_{q \in Q_\mu(\tilde{s}, \tilde{b}, \hat{p})} \mu(q)},$$

$\xi(s, b \mid \hat{p}) = 0$  for all  $(s, b) \notin \cup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})$ . Thus, (B.8) implies

$$\begin{aligned} & \sum_{(s,b) \in \cup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \Pr(s, b) \sum_{q \in Q_\mu(s,b,\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) \hat{\alpha} \cdot \mathbf{1}_{\{r(\hat{\alpha}|s,b,\hat{p})=(Y,Y)\}} \\ & > \sum_{(s,b) \in \cup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \Pr(s, b) \sum_{q \in Q_\mu(s,b,\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) \hat{p} \cdot \mathbf{1}_{\{r(\hat{p}|s,b,\hat{p})=(Y,Y)\}}. \end{aligned}$$

Combined with (B.9), I have

$$\begin{aligned} & \sum_{q \in Q} \mu(q) V(q \circ \alpha, r) \\ &= \sum_{q \in Q_\mu(\hat{p})} \mu(q) V(q \circ \alpha, r) + \sum_{q \notin Q_\mu(\hat{p})} \mu(q) V(q, r) \\ &= \sum_{(s,b) \in \cup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \Pr(s, b) \left[ \sum_{q \in Q_\mu(s,b,\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) \hat{\alpha} \cdot \mathbf{1}_{\{r(\hat{\alpha}|s,b,\hat{p})=(Y,Y)\}} \right. \\ & \quad \left. + \sum_{q \in Q_\mu(\hat{p}) \setminus Q_\mu(s,b,\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) q(s, b) \cdot \mathbf{1}_{\{r(q(s,b)|s,b,q(s,b))=(Y,Y)\}} \right] \\ & \quad + \sum_{(s,b) \notin \cup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \Pr(s, b) \sum_{q \in Q_\mu(\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) q(s, b) \cdot \mathbf{1}_{\{r(q(s,b)|s,b,q(s,b))=(Y,Y)\}} \\ & \quad + \sum_{q \notin Q_\mu(\hat{p})} \mu(q) V(q, r) \\ & > \sum_{(s,b) \in \cup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \Pr(s, b) \left[ \sum_{q \in Q_\mu(s,b,\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) \hat{p} \cdot \mathbf{1}_{\{r(\hat{p}|s,b,\hat{p})=(Y,Y)\}} \right. \\ & \quad \left. + \sum_{q \in Q_\mu(\hat{p}) \setminus Q_\mu(s,b,\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) q(s, b) \cdot \mathbf{1}_{\{r(q(s,b)|s,b,q(s,b))=(Y,Y)\}} \right] \\ & \quad + \sum_{(s,b) \notin \cup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \Pr(s, b) \sum_{q \in Q_\mu(\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) q(s, b) \cdot \mathbf{1}_{\{r(q(s,b)|s,b,q(s,b))=(Y,Y)\}} \\ & \quad + \sum_{q \notin Q_\mu(\hat{p})} \mu(q) V(q, r) \\ &= \sum_{q \in Q_\mu(\hat{p})} \mu(q) V(q, r) + \sum_{q \notin Q_\mu(\hat{p})} \mu(q) V(q, r) \\ &= \sum_{q \in Q} \mu(q) V(q, r). \end{aligned}$$

Hence, (2.1) does not hold.

Next, suppose that (B.7) is not satisfied for  $\mu \in \Delta(Q \times \{r\})$ ; that is, there exists some trader  $i \in \{S, B\}$  of type  $\hat{\theta}_i \in \Theta_i$  and his manipulation  $(\hat{\beta}_i, \hat{\gamma}_i) \in \hat{\Sigma}_i$  such that

$$\sum_{q \in Q} \mu(q) U_i((q, r) \circ (\hat{\beta}_i, \hat{\gamma}_i) \mid \hat{\theta}_i) > \sum_{q \in Q} \mu(q) U_i(q, r \mid \hat{\theta}_i).$$

Consider an ex ante manipulation  $(\beta_i, \gamma_i) \in \Sigma_i$  of trader  $i$  such that

$$\beta_i(\theta_i) = \begin{cases} \hat{\beta}_i & \text{if } \theta_i = \hat{\theta}_i \\ \theta_i & \text{if } \theta_i \neq \hat{\theta}_i \end{cases}$$

and for any  $a_i \in \{Y, N\}$  and  $p, p' \in \mathbb{R}_+$ ,

$$\gamma_i(\theta_i, a_i, p') = \begin{cases} \hat{\gamma}_i(a_i, p') & \text{if } \theta_i = \hat{\theta}_i \\ r_i(p' \mid \theta_i, \theta_j, p) & \text{if } \theta_i \neq \hat{\theta}_i. \end{cases}$$

In other words, trader  $i$  disobeys as prescribed in  $(\hat{\beta}_i, \hat{\gamma}_i)$  only when his type is  $\hat{\theta}_i$ . Then, I have

$$\begin{aligned} & \sum_{q \in Q} \mu(q) U_i((q, r) \circ (\beta_i, \gamma_i)) \\ &= \Pr(\hat{\theta}_i) \sum_{q \in Q} \mu(q) U_i((q, r) \circ (\hat{\beta}_i, \hat{\gamma}_i) \mid \hat{\theta}_i) + (1 - \Pr(\hat{\theta}_i)) \sum_{q \in Q} \mu(q) U_i(q, r \mid \theta_i \neq \hat{\theta}_i) \\ &> \Pr(\hat{\theta}_i) \sum_{q \in Q} \mu(q) U_i(q, r \mid \hat{\theta}_i) + (1 - \Pr(\hat{\theta}_i)) \sum_{q \in Q} \mu(q) U_i(q, r \mid \theta_i \neq \hat{\theta}_i) \\ &= \sum_{q \in Q} \mu(q) U_i(q, r). \end{aligned}$$

Hence, (2.2) does not hold. This completes the proof of the only-if-part.  $\square$

To prove the if-part, suppose that (2.1) is not satisfied for  $\mu \in \Delta(Q \times \{r\})$ ; that is, there exists some manipulation  $\alpha \in \Sigma_I$  by the intermediary such that

$$\sum_{q \in Q} \mu(q) V(q \circ \alpha, r) > \sum_{q \in Q} \mu(q) V(q, r). \quad (\text{B.10})$$

For any  $\alpha \in \Sigma_I$ , let  $P_\alpha$  be the set of prices such that  $\alpha$  prescribes disobeying; that is,

$$P_\alpha = \{p \in \mathbb{R}_+ : \alpha(p) \neq p\}.$$

Note that (B.10) can be written as

$$\begin{aligned}
& \sum_{q \in Q} \mu(q) V(q \circ \alpha, r) \\
&= \sum_{q \notin \bigcup_{p \in P_\alpha} Q_\mu(p)} \mu(q) V(q, r) + \sum_{p \in P_\alpha} \sum_{q \in Q_\mu(p)} \mu(q) V(q \circ \alpha, r) \\
&= \sum_{q \notin \bigcup_{p \in P_\alpha} Q_\mu(p)} \mu(q) V(q, r) + \sum_{p \in P_\alpha} \sum_{q \in Q_\mu(p)} \mu(q) \sum_{(s,b) \in \Theta_q(p)} \Pr(s, b) \cdot (\delta_S + \delta_B) \alpha(p) \cdot \mathbf{1}_{\{r(\alpha(p)|s,b,p)=(Y,Y)\}} \\
&\quad + \sum_{q \in \bigcup_{p \in P_\alpha} Q_\mu(p)} \mu(q) \sum_{(s,b) \notin \bigcup_{p \in P_\alpha} \Theta_q(p)} \Pr(s, b) \cdot (\delta_S + \delta_B) q(s, b) \cdot \mathbf{1}_{\{r(q(s,b)|s,b,q(s,b))=(Y,Y)\}} \\
&> \sum_{q \notin \bigcup_{p \in P_\alpha} Q_\mu(p)} \mu(q) V(q, r) + \sum_{p \in P_\alpha} \sum_{q \in Q_\mu(p)} \mu(q) \sum_{(s,b) \in \Theta_q(p)} \Pr(s, b) \cdot (\delta_S + \delta_B) p \cdot \mathbf{1}_{\{r(p|s,b,p)=(Y,Y)\}} \\
&\quad + \sum_{q \in \bigcup_{p \in P_\alpha} Q_\mu(p)} \mu(q) \sum_{(s,b) \notin \bigcup_{p \in P_\alpha} \Theta_q(p)} \Pr(s, b) \cdot (\delta_S + \delta_B) q(s, b) \cdot \mathbf{1}_{\{r(q(s,b)|s,b,q(s,b))=(Y,Y)\}} \\
&= \sum_{q \notin \bigcup_{p \in P_\alpha} Q_\mu(p)} \mu(q) V(q, r) + \sum_{p \in P_\alpha} \sum_{q \in Q_\mu(p)} \mu(q) V(q, r) \\
&= \sum_{q \in Q} \mu(q) V(q, r).
\end{aligned}$$

Then, there exists at least one recommendation  $\hat{p} \in P_\alpha$  such that

$$\begin{aligned}
& \sum_{q \in Q_\mu(\hat{p})} \mu(q) \sum_{(s,b) \in \Theta_q(\hat{p})} \Pr(s,b) \cdot (\delta_S + \delta_B) \alpha(\hat{p}) \cdot \mathbf{1}_{\{r(\alpha(\hat{p})|s,b,\hat{p})=(Y,Y)\}} \\
& > \sum_{q \in Q_\mu(\hat{p})} \mu(q) \sum_{(s,b) \in \Theta_q(\hat{p})} \Pr(s,b) \cdot (\delta_S + \delta_B) \hat{p} \cdot \mathbf{1}_{\{r(\hat{p}|s,b,\hat{p})=(Y,Y)\}} \\
\iff & \sum_{(s,b) \in \bigcup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \Pr(s,b) \sum_{q \in Q_\mu(s,b,\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) \alpha(\hat{p}) \cdot \mathbf{1}_{\{r(\alpha(\hat{p})|s,b,\hat{p})=(Y,Y)\}} \\
& > \sum_{(s,b) \in \bigcup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \Pr(s,b) \sum_{q \in Q_\mu(s,b,\hat{p})} \mu(q) \cdot (\delta_S + \delta_B) \hat{p} \cdot \mathbf{1}_{\{r(\hat{p}|s,b,\hat{p})=(Y,Y)\}} \\
\iff & \sum_{(s,b) \in \bigcup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \underbrace{\frac{\Pr(s,b) \sum_{q \in Q_\mu(s,b,\hat{p})} \mu(q)}{\sum_{(\tilde{s},\tilde{b}) \in \Theta_S \times \Theta_B} \Pr(\tilde{s},\tilde{b}) \sum_{q \in Q_\mu(\tilde{s},\tilde{b},\hat{p})} \mu(q)}}_{= \xi(s,b|\hat{p})} \cdot (\delta_S + \delta_B) \alpha(\hat{p}) \cdot \mathbf{1}_{\{r(\alpha(\hat{p})|s,b,\hat{p})=(Y,Y)\}} \\
& > \sum_{(s,b) \in \bigcup_{q \in Q_\mu(\hat{p})} \Theta_q(\hat{p})} \underbrace{\frac{\Pr(s,b) \sum_{q \in Q_\mu(s,b,\hat{p})} \mu(q)}{\sum_{(\tilde{s},\tilde{b}) \in \Theta_S \times \Theta_B} \Pr(\tilde{s},\tilde{b}) \sum_{q \in Q_\mu(\tilde{s},\tilde{b},\hat{p})} \mu(q)}}_{= \xi(s,b|\hat{p})} \cdot (\delta_S + \delta_B) \hat{p} \cdot \mathbf{1}_{\{r(\hat{p}|s,b,\hat{p})=(Y,Y)\}} \\
\iff & \sum_{(s,b) \in \Theta_S \times \Theta_B} \xi(s,b|\hat{p}) \cdot (\delta_S + \delta_B) \alpha(\hat{p}) \cdot \mathbf{1}_{\{r(\alpha(\hat{p})|s,b,\hat{p})=(Y,Y)\}} \\
& > \sum_{(s,b) \in \Theta_S \times \Theta_B} \xi(s,b|\hat{p}) \cdot (\delta_S + \delta_B) \hat{p} \cdot \mathbf{1}_{\{r(\hat{p}|s,b,\hat{p})=(Y,Y)\}} \\
\iff & V(\mu \circ \alpha(\hat{p}) | \hat{p}) > V(\mu | \hat{p}).
\end{aligned}$$

Hence, (B.6) does not hold.

Next, suppose that (2.2) is not satisfied for  $\mu \in \Delta(Q \times \{r\})$ ; that is, there exists some trader  $i \in \{S, B\}$  and his manipulation  $(\beta_i, \gamma_i) \in \Sigma_i$  such that

$$\sum_{q \in Q} \mu(q) U_i((q, r) \circ (\beta_i, \gamma_i)) > \sum_{q \in Q} \mu(q) U_i(q, r).$$

Note that, as seen in the proof of the only-if-part, trader  $i$ 's ex ante expected payoff is a weighted average of his interim expected payoffs. Thus, there exists at least one type  $\hat{\theta}_i \in \Theta_i$  who gets a strictly higher interim expected payoff from this manipulation. Let  $(\hat{\beta}_i, \hat{\gamma}_i) \in \hat{\Sigma}_i$  denote the

interim manipulation by  $\hat{\theta}_i$  obtained from  $(\beta_i, \gamma_i)$ . I must have

$$\sum_{q \in Q} \mu(q) U_i((q, r) \circ (\hat{\beta}_i, \hat{\gamma}_i) \mid \hat{\theta}_i) > \sum_{q \in Q} \mu(q) U_i(q, r \mid \hat{\theta}_i).$$

Hence, (B.7) does not hold. This completes the proof.

## C Alternative Assumptions on the Intermediary

### C.1 Unbiased intermediary

#### C.1.1 With commitment and enforcement power

Assuming binary valuations as in the present paper, Matsuo (1989) studies the mechanism design problem for bilateral trade à la Myerson and Satterthwaite (1983). The principal in Matsuo (1989) can be viewed as an unbiased intermediary with the commitment and enforcement power in the sense that (i) her objective is to maximize the expected social surplus, which does not depend on trading price, and (ii) she offers a *mechanism* for which *ex ante* individual rationality constraints must be satisfied. He shows that a Bayesian incentive compatible, individually rational, and ex post efficient mechanism exists if and only if

$$\pi_S \pi_B b_H + (1 - \pi_S) b_L \geq \pi_B s_H + (1 - \pi_S)(1 - \pi_B) s_L. \quad (\text{C.1})$$

As I show in Appendix C.1, condition (C.1) is essentially the same as the necessary and sufficient condition for the existence of ex post efficient pure acceptable CE when the intermediary is unbiased. Thus, when it comes to achieving ex post efficiency, the lack of commitment and enforcement power has no bite as long as the intermediary is unbiased.

#### C.1.2 Without commitment and enforcement power

Assume the following payoff specification: if the traders  $(s, b)$  trade at price  $p$ ,

- the seller gets  $p - s - f_S$ ;
- the buyer gets  $b - p - f_B$ ;

- the intermediary gets  $f \equiv f_S + f_B$ ,

where  $f_S, f_B \in \mathbb{R}_+$  and  $f > 0$ . That is, the intermediary gets a fixed commission  $f_i$  from trader  $i \in \{S, B\}$  if trade occurs. The intermediary with this payoff function is said *unbiased* since his payoff no longer depends on the sale price. Note that the seller  $s$  gets a nonnegative payoff by accepting  $p$  if and only if  $p \geq s + f_S$  and that the buyer  $b$  does so if and only if  $p \leq b - f_B$ . Assume  $f \leq \min\{b_L - s_L, b_H - s_H\}$  so that there is room for achieving ex post efficiency. I can show an analog of Lemma 1, a necessary condition for acceptable CE in the unbiased case.

**Lemma 6.** *Suppose that the intermediary is unbiased. A mediation plan  $\mu \in \Delta(Q \times \{r^A\})$  is an acceptable CE only if it recommends the intermediary to offer a mutually acceptable price whenever there is gain from trade. That is, for all  $q \in \text{supp}(\mu)$ ,*

$$q(s, b) \in [s + f_S, b - f_B] \text{ for all } (s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\} \quad (\text{C.2})$$

The proof is almost the same as that of Lemma 1 thus omitted. As to the recommendation to reports  $(s_H, b_L)$ , a similar insight in the biased case holds; I can restrict my attention to the price-recommendations in  $\bar{Q} \subset Q$  defined below:

$$\bar{Q} = \{q \in Q : q \text{ satisfies (C.2) and } q(s_H, b_L) \in (b_H - f_B, +\infty)\}.$$

Consider an arbitrary mediation plan  $\mu \in \Delta(Q \times \{r^A\})$  that satisfies (C.2) for all  $q \in \text{supp}(\mu)$ . For any  $q \in \text{supp}(\mu)$ , let  $\bar{q} \in \bar{Q}$  be an arbitrary price recommendation such that  $\bar{q}(s, b) = q(s, b)$  for all  $(s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\}$ . Construct a mediation plan  $\bar{\mu} \in \Delta(\bar{Q} \times \{r^A\})$  by letting  $\bar{\mu}(\bar{q}) = \mu(q)$  for all  $q \in \text{supp}(\mu)$ . Here is an analog of Lemma 2.

**Lemma 7.** *Suppose that the intermediary is unbiased. For any mediation plan  $\mu \in \Delta(Q \times \{r^A\})$ , if  $\mu$  is an acceptable CE, so is  $\bar{\mu}$ .*

*Proof.* The proof is almost the same as that of Lemma 2; one can show that the players' IC constraints are (weakly) more stringent in  $\mu$  than in  $\bar{\mu}$ . In particular, the high-type buyer and the low-type seller have weakly more room for profitable manipulations.  $\square$

Since  $\mu$  and  $\bar{\mu}$  induce the same distribution over the bargaining outcome, it is without loss of generality to focus on  $\bar{\mu}$ . Using Lemma 7, I can establish a necessary and sufficient condition for the existence of ex post efficient pure acceptable CE.<sup>26</sup>

**Proposition 3.** *If the intermediary is unbiased, an ex post efficient pure acceptable CE exists if and only if*

$$\pi_S \pi_B b_H + (1 - \pi_S) b_L \geq \pi_B s_H + (1 - \pi_S)(1 - \pi_B) s_L + [1 - \pi_S(1 - \pi_B)] f. \quad (\text{C.3})$$

*Proof.* Given Matsuo (1989), the only-if-part is trivial; if ex post efficiency cannot be achieved even *with* commitment and enforcement, not to mention the case *without* them.

To prove the if-part, consider a pure mediation plan  $(q, r^A) \in \bar{Q} \times \{r^A\}$  such that  $q(s_H, b_H) = s_H + f_S$ ,  $q(s_L, b_L) = b_L - f_B$ , and  $q(s_L, b_H) = p$  for some  $p \in [b_L - f_B, s_H + f_S]$ . It is easy to see that the high-type seller and the low-type buyer have no profitable manipulation. The low-type seller has no incentive for manipulation if

$$\begin{aligned} & \pi_B(p - s_L - f_S) + (1 - \pi_B)(b_L - f_B - s_L - f_S) \geq \pi_B(s_H + f_S - s_L - f_S) \\ \iff & \pi_B \leq \frac{b_L - s_L - f}{s_H - p + b_L - s_L - f_B}. \end{aligned} \quad (\text{C.4})$$

The high-type buyer has no incentive for manipulation if

$$\begin{aligned} & \pi_S(b_H - s_H - f_S - f_B) + (1 - \pi_S)(b_H - p - f_B) \geq (1 - \pi_S)(b_H - b_L + f_B - f_B) \\ \iff & \pi_S \geq \frac{p - b_L + f_B}{b_H - s_H + p - b_L - f_S}. \end{aligned} \quad (\text{C.5})$$

Solving (C.4) and (C.5) with respect to  $p$ , I obtain

$$p \in \left[ s_H - \frac{1 - \pi_B}{\pi_B}(b_L - s_L) + \frac{1}{\pi_B}[f_S + (1 - \pi_B)f_B], b_L + \frac{\pi_S}{1 - \pi_S}(b_H - s_H) - \frac{1}{1 - \pi_S}[\pi_S f_S + f_B] \right].$$

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<sup>26</sup>Since Matsuo (1989) focuses on deterministic mechanisms, I restrict my attention to *pure* acceptable CE.

Since

$$s_H - \frac{1 - \pi_B}{\pi_B}(b_L - s_L) + \frac{1}{\pi_B}[f_S + (1 - \pi_B)f_B] \leq s_H + f_S \iff f \leq b_L - s_L$$

and

$$b_L + \frac{\pi_S}{1 - \pi_S}(b_H - s_H) - \frac{1}{1 - \pi_S}[\pi_S f_S + f_B] \geq b_L - f_B \iff f \leq b_H - s_H,$$

such  $p$  exists if and only if

$$\begin{aligned} s_H - \frac{1 - \pi_B}{\pi_B}(b_L - s_L) + \frac{1}{\pi_B}[f_S + (1 - \pi_B)f_B] &\leq b_L + \frac{\pi_S}{1 - \pi_S}(b_H - s_H) - \frac{1}{1 - \pi_S}[\pi_S f_S + f_B] \\ \iff \pi_S \pi_B b_H + (1 - \pi_S)b_L &\geq \pi_B s_H + (1 - \pi_S)(1 - \pi_B)s_L + [1 - \pi_S(1 - \pi_B)]f. \end{aligned}$$

Next, consider the intermediary's incentive.

- If  $p \in (b_L - f_B, s_H + f_S)$ , for each  $(s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\}$ , the intermediary learns that the traders' are  $(s, b)$  upon recommended  $q(s, b)$ . Hence, his expected payoff is  $f$  if he offers  $p \in [s + f_S, b - f_B]$  and 0 otherwise.
- If  $p = b_L - f_B$ :
  - upon recommended  $s_H + f_S$ , the intermediary learns that the traders are  $(s_H, b_H)$ . Hence, his expected payoff is  $f$  if he offers  $p \in [s_H + f_S, b_H - f_B]$  and 0 otherwise.
  - If  $b_L - f_B$  is recommended, the intermediary learns that the traders are either  $(s_L, b_H)$  or  $(s_L, b_L)$ . Hence, his expected payoff is  $f$  if he offers  $p \in [s_L + f_S, b_L - f_B]$ ,  $\pi_B f$  if he offers  $p \in (b_L - f_B, b_H - f_B]$ , and 0 otherwise.
- If  $p = s_H + f_S$ :
  - upon recommended  $b_L - f_B$ , the intermediary learns that the traders are  $(s_L, b_L)$ . Hence, his expected payoff is  $f$  if he offers  $p \in [s_L + f_S, b_L - f_B]$  and 0 otherwise.
  - upon recommended  $s_H + f_S$ , the intermediary learns that the traders are either  $(s_H, b_H)$  or  $(s_L, b_H)$ . Hence, his expected payoff is  $f$  if he offers  $p \in [s_H + f_S, b_H - f_B]$ ,  $(1 - \pi_S)f$  if he offers  $p \in [s_L + f_S, s_H + f_S)$ , and 0 otherwise.

- Regardless of the value of  $p$ , upon recommended  $q(s_H, b_L)$ , the intermediary learns that the traders are  $(s_H, b_L)$ . Hence, his expected payoff is 0, no matter what price he offers.

Thus, the intermediary has no profitable manipulation. This completes the proof.  $\square$

Condition (C.3) is essentially the same as (C.1). Indeed, if I modify the traders' payoffs (i) in Matsuo (1989) so that they pay the fixed commissions; or (ii) in the present model so that the traders do not pay the commissions (as in Section 5.2.1), the two conditions coincide. Note that (C.3) can be also written as

$$\begin{aligned}
& \underbrace{\pi_S \pi_B (b_H - s_H) + (1 - \pi_S) \pi_B (b_H - s_L) + (1 - \pi_S)(1 - \pi_B)(b_L - s_L)}_{\text{The expected trade surplus}} \\
& \geq \underbrace{(1 - \pi_S) \pi_B (b_H - b_L + s_H - s_L)}_{\text{The expected incentive cost}} + \underbrace{[1 - \pi_S(1 - \pi_B)]f}_{\text{The expected commissions}}.
\end{aligned} \tag{C.6}$$

That is, ex post efficiency can be achieved if and only if the expected trade surplus exceeds the expected total cost of trade. To see why the first term on the right side represents the incentive cost, recall that the only relevant IC constraints are those of the low-type seller and the high-type buyer, (C.4) and (C.5). If the intermediary could offer a different price to each trader, these constraints would have no cost; if he could offer  $s_H + f_S$  to the seller and  $b_L - f_B$  to the buyer, the constraints would be reduced to  $\pi_B \leq 1$  and  $\pi_S \geq 0$ . In addition, this would yield the expected revenue of  $(1 - \pi_S) \pi_B (b_H - b_L + s_H - s_L)$ . Hence, this term can be interpreted as the incentive cost.

The relationship between (C.3) and its counterpart in the biased case is worth mentioning. As shown in Proposition 1, if the intermediary is biased, an ex post efficient acceptable CE exists if and only if  $\pi_B \leq \overline{\pi_B} = \frac{b_L - h s_L}{b_H - h s_L} \Leftrightarrow 1 - \pi_B \geq \frac{b_H - b_L}{b_H - h s_L}$ , where  $h = \frac{1 + \delta_B}{1 - \delta_S} \leq \min\left\{\frac{b_L}{s_L}, \frac{b_H}{s_H}\right\}$ . Note that

$$1 - \pi_B \geq \frac{b_H - b_L}{b_H - h s_L} \Leftrightarrow \pi_S \pi_B b_H + (1 - \pi_S) b_L \geq \pi_B b_H + (1 - \pi_S)(1 - \pi_B) h s_L.$$

Hence,  $\pi_B \leq \overline{\pi_B}$  implies (C.3) if

$$\begin{aligned} \pi_B b_H + (1 - \pi_S)(1 - \pi_B) h s_L &\geq \pi_B s_H + (1 - \pi_S)(1 - \pi_B) s_L + [1 - \pi_S(1 - \pi_B)]f \\ \iff \pi_B(b_H - s_H) + (1 - \pi_S)(1 - \pi_B)(h - 1)s_L &\geq [1 - \pi_S(1 - \pi_B)]f. \end{aligned} \quad (\text{C.7})$$

On the one hand, if  $f$  is sufficiently small, (C.7) holds for any  $h$ ; if the commission for an unbiased intermediary is sufficiently low, even a tiny bias is detrimental to ex post efficiency. On the other hand, if  $f$  is large, (C.7) does not hold for  $h$  small; if the commission for an unbiased intermediary is high but the commission rates for a biased intermediary are small, in some cases, ex post efficiency can be achieved with the biased intermediary but not with the unbiased one. In other words, a biased intermediary can perform better than an unbiased intermediary.<sup>27</sup>

## C.2 Without intermediary

In this appendix, I consider another mediated bargaining game, the *mediated seller-offer bargaining game*, in which there is no intermediary, and the seller instead makes an offer. I assume that if the traders  $(s, b)$  trade at price  $p$ , the seller gets  $p - s$ , and the buyer gets  $b - p$ . Let  $\hat{r}: \Theta_S \times \Theta_B \times \mathbb{R}_+^2 \rightarrow \{Y, N\}$  be a response-recommendation to the buyer and  $\hat{R}$  be the set of all such recommendations. A mediation plan is a probability distribution over  $Q \times \hat{R}$ . The timing of the game is as follows:

1. The mediator publicly commits to a mediation plan  $\mu \in \Delta(Q \times \hat{R})$  she uses;
2. The seller and the buyer privately observe the realizations of their types;
3. The traders privately report their types  $\tilde{s} \in \Theta_S$  and  $\tilde{b} \in \Theta_B$  to the mediator;
4. The mediator privately picks up a pure mediation plan  $(q, \hat{r}) \in Q \times \hat{R}$  with probability  $\mu(q, \hat{r})$  and then privately recommends a price  $p = q(\tilde{s}, \tilde{b})$  to the seller;
5. The seller offers a price  $\tilde{p} \in \mathbb{R}_+$  to the buyer, and the mediator observes it;

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<sup>27</sup>This observation crucially hinges on the fact that the traders pay (varying or fixed) commissions. If the intermediary is solely interested in maximizing the probability of trade and does not take commissions from the traders ( $f_S = f_B = 0$ ), (C.7) always holds; that is, having a bias is always detrimental to ex post efficiency.

6. The mediator privately recommends the buyer's response  $\hat{r}(\tilde{p} \mid s, b, p)$ ;
7. The buyer responds to the offer  $\tilde{p}$  by either acceptance ( $Y$ ) or rejection ( $N$ ). If he accepts it, trade occurs at the price  $\tilde{p}$ , and the payoffs are realized; if he rejects it, no trade occurs, and both players get a payoff of 0.

Codomination correspondence can be defined in a similar manner as before, and I can show that no action is codominated in this game too, which implies any CE is SCE. Let  $\hat{r}^A \in \hat{R}$  be a response-recommendation such that for all  $p \in \mathbb{R}_+$  and all  $(s, b) \in \Theta_S \times \Theta_B$ ,

$$\hat{r}^A(\tilde{p} \mid s, b, p) = \begin{cases} Y & \text{if } \tilde{p} \in [0, b] \\ N & \text{if } \tilde{p} \in (b, +\infty) \end{cases}.$$

A CE  $\mu \in \Delta(Q \times \hat{R})$  is *acceptable* if  $\mu(Q \times \{\hat{r}^A\}) = 1$ .

I first provide a necessary condition for acceptable CE.

**Lemma 8.** *A mediation plan  $\mu \in \Delta(Q \times \{\hat{r}^A\})$  is an acceptable CE only if it always recommends  $b_H$  to the traders  $(s_H, b_H)$  and recommends either  $b_L$  or  $b_H$  to the low-type seller  $s_L$ . That is, for all  $q \in \text{supp}(\mu)$ ,*

$$\begin{aligned} q(s_H, b_H) &= b_H \\ q(s_L, b) &\in \{b_L, b_H\} \text{ for all } b \in \Theta_B. \end{aligned} \tag{C.8}$$

*Proof.* I show the contraposition. Consider an arbitrary mediation plan  $\mu \in \Delta(Q \times \{\hat{r}^A\})$  that does not satisfy (C.8). I show that the seller has a profitable manipulation. Note that if the seller reports  $\tilde{s}$  and recommended  $p$ , his posterior belief that the buyer is of high type is given by

$$\xi(b_H \mid \tilde{s}, p) = \frac{\pi_B \sum_{q \in \{q' \in Q: q'(\tilde{s}, b_H)=p\}} \mu(q)}{\pi_B \sum_{q \in \{q' \in Q: q'(\tilde{s}, b_H)=p\}} \mu(q) + (1 - \pi_B) \sum_{q \in \{q' \in Q: q'(\tilde{s}, b_L)=p\}} \mu(q)}$$

- Suppose  $q(s_H, b_H) = p \neq b_H$ . If the high-type seller honestly reports his type and recommended  $p$ , his posterior is  $\xi(b_H \mid s_H, p) > 0$ . If he offers  $\tilde{p}$ , his expected payoff is

$$- \xi(b_H \mid s_H, p)(\tilde{p} - s_H) \geq 0 \text{ if } \tilde{p} \in [s_H, b_H];$$

- nonpositive if  $\tilde{p} \notin [s_H, b_H]$ .

Hence, he would gain by offering  $b_H$  when recommended  $p$ .

- Suppose  $q(s_L, b) = p \notin \{b_L, b_H\}$  for some  $b \in \Theta_B$ . If the low-type seller honestly reports his type and recommended  $p$ , his posterior is  $\xi(b_H | s_L, p) \geq 0$ . If he offers  $\tilde{p}$ , his expected payoff is

- $\xi(b_H | s_L, p)(\tilde{p} - s_L) \geq 0$  if  $\tilde{p} \in (b_L, b_H]$ ;
- $\tilde{p} - s_L \geq 0$  if  $\tilde{p} \in [s_L, b_L]$ ;
- nonpositive if  $\tilde{p} \notin [s_L, \overline{p_{b_H}}]$ .

Hence, he would gain by offering either  $b_L$  or  $b_H$  when recommended  $p$ .

□

Lemma 9 below implies that it is without loss of generality to focus on mediation plans in  $\Delta(\hat{Q}_x, \{r^A\})$ , where

$$\hat{Q}_x = \{q \in Q : q \text{ satisfies (C.8) and } q(s_H, b_L) = b_H\}.$$

The intuition is similar to that in Section 3. For any mediation plan  $\mu \in \Delta(Q \times \{\hat{r}^A\})$  that satisfies (C.8) for all  $q \in \text{supp}(\mu)$ , I can construct another mediation plan  $\mu_x \in \Delta(\hat{Q}_x \times \{\hat{r}^A\})$  that has weakly less stringent IC constraints than  $\mu$  and induces the same distribution over the bargaining outcomes if the players are honest and obedient to the mediator. For any  $q$  in the support of such  $\mu$ , let  $x_q \in \hat{Q}_x$  be the price-recommendation such that  $x_q(s, b) = q(s, b)$  for all  $(s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\}$ . That is,  $x_q$  recommends the same prices as  $q$  for all reports  $(s, b) \in \Theta_S \times \Theta_B \setminus \{(s_H, b_L)\}$  but recommends  $b_H$  for reports  $(s_H, b_L)$ . The mediation plan  $\mu_x$  can be constructed by letting  $\mu_x(x_q) = \mu(q)$  for all  $q \in \text{supp}(\mu)$ .

**Lemma 9.** *For any mediation plan  $\mu \in \Delta(Q \times \{\hat{r}^A\})$ , if  $\mu$  is an acceptable CE, so is  $\mu_x$ .*

*Proof.* Consider an arbitrary mediation plan  $\mu \in \Delta(Q \times \{\hat{r}^A\})$  that satisfies (C.8) for all  $q \in \text{supp}(\mu)$ . By construction, both traders get the same expected payoffs in  $\mu$  and  $\mu_x$  if they are

honest and obedient to the mediator. Hence, it suffices to show that  $\mu$  has weakly more room for profitable manipulations for both players than  $\mu_x$ . As in Section 3, let  $x_{HL}$ ,  $x_{LH}$ , and  $x_{LL}$  denote the total probabilities that price  $b_H$  is recommended for each pair of reports in  $\mu$ . By a similar argument as in the proof of Lemma 1, the IC constraints for both types of buyers are weakly more stringent in  $\mu$  than in  $\mu_x$ . Since the high-type seller finds it optimal to offer  $b_H$  whatever belief about the buyer's type he has, his IC constraints are the same in  $\mu$  and  $\mu_x$ . Consider the low-type seller.

- Suppose  $x_{HL} = 0$ . If he misreports his type, the recommended price is conclusive about the buyer's type. Hence, if he is recommended  $b_H$ , he would follow the recommendation and get  $b_H - s_L$ ; if he is recommended  $p \neq b_H$ , he would offer  $b_L$  and get  $b_L - s_L$ . Thus, his expected payoff from honest and obedient behavior must be at least as large as

$$\pi_B(b_H - s_L) + (1 - \pi_B)(b_L - s_L).$$

- Suppose  $x_{HL} \in (0, 1)$ . If he misreports his type, the recommendation  $p \neq b_H$  is conclusive about the buyer's type, but  $b_H$  is not. If he is recommended  $b_H$ , his expected payoff is

$$\begin{aligned} & - \frac{\pi_B}{\pi_B + (1 - \pi_B)x_{HL}}(b_H - s_L) \text{ if he offers } b_H; \\ & - b_L - s_L \text{ if he offers } b_L. \end{aligned}$$

If he is recommended  $p \neq b_H$ , he would offer  $b_L$  and get  $b_L - s_L$ . Thus, his expected payoff from honest and obedient behavior must be at least as large as

$$\begin{aligned} & [\pi_B + (1 - \pi_B)x_{HL}] \frac{\pi_B}{\pi_B + (1 - \pi_B)x_{HL}}(b_H - s_L) + (1 - \pi_B)(1 - x_{HL})(b_L - s_L) \\ & = \pi_B(b_H - s_L) + (1 - \pi_B)(1 - x_{HL})(b_L - s_L). \end{aligned}$$

Since this expected payoff is decreasing in  $x_{HL}$ , the IC constraint in  $\mu$  is at least as stringent as that in  $\mu_x$ .

The above argument suffices for the proof, but I spell out the players' IC constraints in  $\mu_x$ . The buyer's IC constraints are the same as those in the game with the intermediary; the low-type

buyer's constraint is trivially satisfied, and the high-type buyer's one is given by

$$x_{LL} \geq x_{LH}. \quad (\text{C.9})$$

The high-type seller's IC constraint is also trivially satisfied. Finally, consider the low-type seller. Suppose that he has honestly reported his type.

- He follows the recommendation  $b_H$  if

$$\begin{aligned} \frac{\pi_B x_{LH}}{\pi_B x_{LH} + (1 - \pi_B) x_{LL}} (b_H - s_L) &\geq b_L - s_L \\ \iff \pi_B (b_H - b_L) x_{LH} - (1 - \pi_B) (b_L - s_L) x_{LL} &\geq 0. \end{aligned} \quad (\text{C.10})$$

- He follows the recommendation  $b_L$  if

$$\begin{aligned} b_L - s_L &\geq \frac{\pi_B (1 - x_{LH})}{\pi_B (1 - x_{LH}) + (1 - \pi_B) (1 - x_{LL})} (b_H - s_L) \\ \iff (1 - \pi_B) (b_L - s_L) (1 - x_{LL}) &\geq \pi_B (b_H - b_L) (1 - x_{LH}) \\ \iff \pi_B (b_H - b_L) x_{LH} - (1 - \pi_B) (b_L - s_L) x_{LL} &\geq \pi_B (b_H - b_L) - (1 - \pi_B) (b_L - s_L). \end{aligned} \quad (\text{C.11})$$

If he misreports his type, he is always recommended  $b_H$ ; hence, no new information about the buyer's type. Thus, the best possible deviation is offering either  $b_H$  or  $b_L$ . He has no incentive to misreport his type if

$$\begin{aligned} \pi_B x_{LH} (b_H - s_L) + [\pi_B (1 - x_{LH}) + (1 - \pi_B) (1 - x_{LL})] (b_L - s_L) \\ \geq \max\{\pi_B (b_H - s_L), b_L - s_L\}. \end{aligned}$$

Note that this condition is reduced to (C.11) if  $\pi_B (b_H - s_L) \geq b_L - s_L$  and to (C.10) if  $\pi_B (b_H - s_L) < b_L - s_L$ . Thus, a mediation plan is an acceptable CE if it satisfies (C.9), (C.10), and (C.11).  $\square$

The definition of ex post efficiency implies that  $\mu \in \Delta(Q \times \{\hat{r}^A\})$  is ex post efficient only

if  $x_{LL} = 0$ . Then, the high-type buyer's IC constraint (C.9) requires  $x_{LH} = 0$ . Thus, the only candidate for ex post efficient acceptable CE is  $(\hat{q}, \hat{r}^A)$  such that  $\hat{q}(s_H, b) = b_H$  and  $\hat{q}(s_L, b) = b_L$  for all  $b \in \Theta_B$ .

**Proposition 4.** *In the mediated seller-offer bargaining game, there exists an ex post efficient acceptable CE if and only if the prior probability  $\pi_B$  of the high-type buyer satisfies  $\pi_B \leq \frac{b_L - s_L}{b_H - s_L}$ .*

*Proof.* Consider the abovementioned mediation plan  $(\hat{q}, \hat{r}^A)$ . The only nontrivial IC constraint is (C.11), which is reduced to  $\pi_B \leq \frac{b_L - s_L}{b_H - s_L}$ . Thus,  $(\hat{q}, \hat{r}^A)$  is an acceptable CE if and only if  $\pi_B \leq \frac{b_L - s_L}{b_H - s_L}$ . Since  $(\hat{q}, \hat{r}^A)$  is the only candidate for ex post efficient acceptable CE, this also proves the only if part.  $\square$

Since  $\frac{b_L - s_L}{b_H - s_L} \geq \overline{\pi_B}$ , if the traders themselves cannot achieve ex post efficiency, then a biased intermediary cannot achieve it either. More importantly, in some cases, ex post efficiency can be achieved without an intermediary but not with a biased one. In this sense, a biased intermediary is not just ineffective but may do harm in the first-best case.

Next, consider the SB case. Lemma 8 shows that there is no acceptable CE that recommends the high-type seller to offer  $b_L$  with positive probability. Such a mediation plan is indeed one of the SB equilibria in the biased intermediary case.<sup>28</sup> Then, intuition tells us that the seller always offers  $b_H$  in the SB equilibrium of the mediated seller-offer bargaining game. Proposition 5 shows that this is correct.

**Proposition 5.** *Suppose  $\pi_B > \frac{b_L - s_L}{b_H - s_L}$ . There is a unique acceptable CE, where the seller always offers price  $b_H$ , and the traders  $(s_L, b_L)$  do not trade with probability 1.*

*Proof.* Suppose  $\pi_B > \frac{b_L - s_L}{b_H - s_L}$ . I show that  $(x_{LH}, x_{LL}) = (1, 1)$  is the unique pair that satisfies the IC constraints. Since (C.11) implies (C.10) when  $\pi_B > \frac{b_L - s_L}{b_H - s_L}$ , the relevant IC constraints are (C.9) and (C.11). As (C.11) becomes more stringent as  $x_{LL}$  increases, given (C.9), it is least stringent at  $x_{LL} = x_{LH}$ . Substituting this into (C.11), I obtain

$$(1 - x_{LL})[(1 - \pi_B)(b_L - s_L) - \pi_B(b_H - b_L)] \geq 0.$$

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<sup>28</sup>Strictly speaking, price  $\overline{p_{b_L}}$  is recommended with positive probability.

Since the inside of the bracket is negative, this condition is satisfied if and only if  $x_{LL} = x_{LH} = 1$ . □

This proposition implies that in the SB case, if the intermediary's bias is not so large ( $h \leq \bar{h}$ ), having a biased intermediary leads to a higher expected trade surplus than having no intermediary. In other words, even a biased intermediary is useful in the SB case.

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